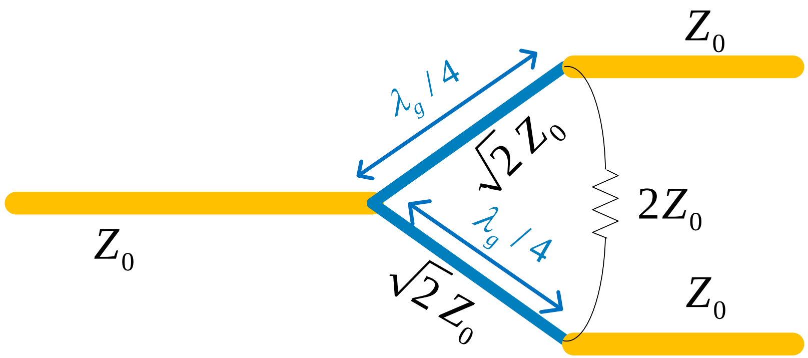
# Microwave Engineering
# EC 401
# DoECE
# Sardar Vallabhbhai National Institute Of Technplogy

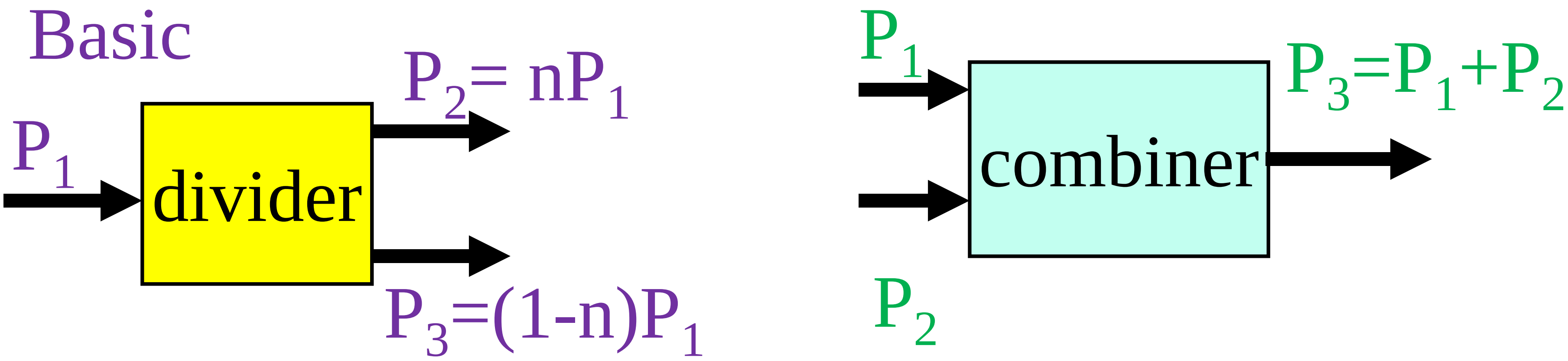
# Module 4
# Power Dividers and Circulators



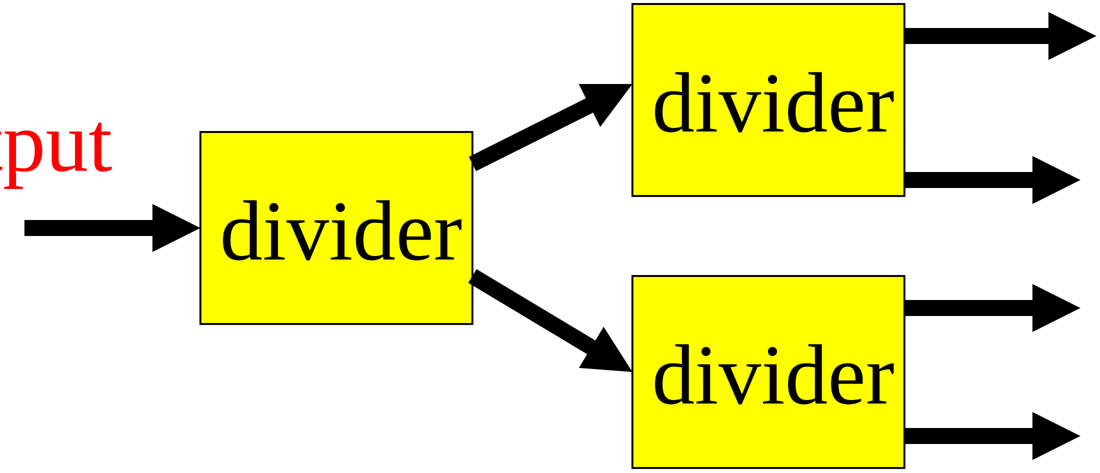

Power Dividers

# Power Divider and Combiner/Coupler

Basic




Divide into 4 output




- ❖ A power divider is used to split a signal.
- ❖ A coupler is used to combine a signal.

# LOSSLESS NETWORK

For lossless n-network,

**total input power = total output power**.

Thus $$\underbrace{\sum_{i=1}^{n} a_i a_i^*}_{a^t a^*} = \underbrace{\sum_{i=1}^{n} b_i b_i^*}_{b^t b^*}$$

Where a and b are the amplitude of the signal

Putting in matrix form $a^t a^* = b^t b^* = a^t S^t S^* a^*$

Note that $b^t = a^t S^t$ and $b^* = S^* a^*$

Thus $a^t (U - S^t S^*) a^* = 0$

This implies that $S^t S^* = U$

# RECIPROCAL AND SYMMETRICAL NETWORK

U is called unitary matrix $\quad [U] = [U]^t$

In summation form $\quad \displaystyle\sum_{k=1}^{n} S_{ki} S_{kj}^{*} = \left.\begin{matrix} 1 & for\, i = j \\ 0 & for\, i \neq j \end{matrix}\right\}$

For reciprocal network

$[Z] = [Z]^t \quad \Rightarrow \quad$ Since [Z] is symmetry

Thus it can be shown that $\quad [S] = [S]^t$

# Properties of the *S* Matrix

For reciprocal networks (networks containing only reciprocal materials), the *S*-matrix is symmetric:

$$S_{ij} = S_{ji} \qquad i \neq j$$

$$\Rightarrow [S] = [S]^T$$

**Example of a nonreciprocal material: a biased ferrite**

# EXAMPLE

A certain two-port network is measured and the following scattering matrix is obtained:

$$[S] = \begin{bmatrix} 0.1\angle 0^o & 0.8\angle 90^o \\ 0.8\angle 90^o & 0.2\angle 0^o \end{bmatrix}$$

- From the data , determine whether the network is reciprocal or lossless.
- If a short circuit is placed on port 2, what will be the resulting return loss at port 1?

# EXAMPLE

***Solution:***

$$[S] = \begin{bmatrix} 0.1\angle 0^o & 0.8\angle 90^o \\ 0.8\angle 90^o & 0.2\angle 0^o \end{bmatrix}$$

Since [S] is symmetry, the network is reciprocal.

To be lossless, the S parameters must satisfy

$$\sum_{k=1}^{n} S_{ki} S_{kj}^{*} = \left.\begin{matrix} 1 & for\, i = j \\ 0 & for\, i \neq j \end{matrix}\right\}$$

For i=j

$|S_{11}|^2 + |S_{12}|^2 = (0.1)^2 + (0.8)^2 = 0.65 \neq 1$

Thus, it is not a lossless network.

# EXAMPLE (CONTD.)

Reflected power at port 1 and port 2:

$a_2 = -b_2$ for port 2 being short circuited

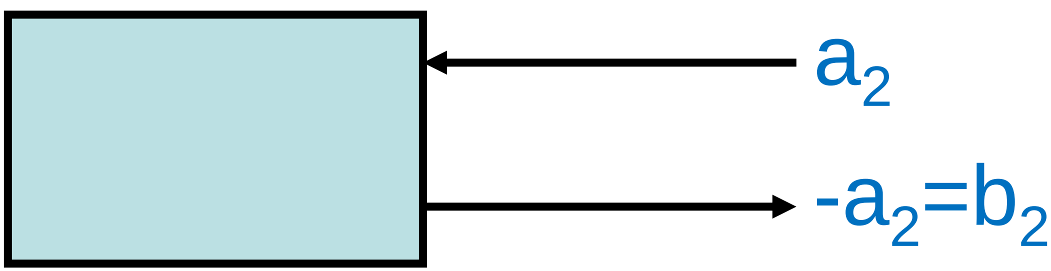
$b_1 = S_{11}a_1 + S_{12}a_2 = S_{11}a_1 - S_{12}b_2$ (1)

$b_2 = S_{21}a_1 + S_{22}a_2 = S_{21}a_1 - S_{22}b_2$ (2)

From (2) we have

$$b_2 = \frac{S_{21}}{1+S_{22}} a_1 \qquad (3)$$

Short at port 2



# EXAMPLE (CONTD.)

Dividing (1) by $a_1$ and substitute the result in (3) ,we have

$$\rho = \frac{b_1}{a_1} = S_{11} - S_{12}\frac{b_2}{a_1} = S_{11} - \frac{S_{12}S_{21}}{1+S_{22}}$$

$$= 0.1 - \frac{(j0.8)(j0.8)}{1+0.2} = 0.633$$

Return loss

$$-20\log\rho = -20\log(0.633) = 3.97\,dB$$

# S-parameter For Power Divider/Coupler

Generally $$[S]=\begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix}$$

For reciprocal and lossless network

$$\boxed{\sum_{k=1}^{N} S_{ki}S_{ki}^{*} = 1}$$

$$\boxed{\sum_{k=1}^{N} S_{ki}S_{kj}^{*} = 0 \qquad for\ i \neq j}$$

$$|S_{11}|^2 + |S_{12}|^2 + |S_{13}|^2 = 1$$

Row 1x row 2 $\rightarrow S_{11}S_{21}^{*} + S_{12}S_{22}^{*} + S_{13}S_{23}^{*} = 0$

$$|S_{21}|^2 + |S_{22}|^2 + |S_{23}|^2 = 1$$

Row 2x row 3 $\rightarrow S_{21}S_{31}^{*} + S_{22}S_{32}^{*} + S_{23}S_{33}^{*} = 0$

$$|S_{31}|^2 + |S_{32}|^2 + |S_{33}|^2 = 1$$

Row 1x row 3 $\rightarrow S_{11}S_{31}^{*} + S_{12}S_{32}^{*} + S_{13}S_{33}^{*} = 0$

# Three Port Networks (cont.)

If all ports are matched properly , then $S_{ii} = 0$

For Reciprocal network

$$[S] = \begin{bmatrix} 0 & S_{12} & S_{13} \\ S_{12} & 0 & S_{23} \\ S_{13} & S_{23} & 0 \end{bmatrix}$$

For lossless network, must satisfy unitary condition

$$|S_{12}|^2 + |S_{13}|^2 = 1 \qquad S_{13}^* S_{23} = 0$$

$$|S_{12}|^2 + |S_{23}|^2 = 1 \qquad S_{23}^* S_{12} = 0$$

$$|S_{13}|^2 + |S_{23}|^2 = 1 \qquad S_{12}^* S_{13} = 0$$

- Two of ($S_{12}$, $S_{13}$, $S_{23}$) must be zero but it is not consistent.
- If $S_{12}=S_{13}= 0$, then $S_{23}$ should equal to 1 and the first equation will not equal to 1.
- This is invalid.

Implying that a three-port network cannot be simultaneously lossless, reciprocal, and matched at all ports.

# Nonreciprocal Network (Apply For Circulator)

For Nonreciprocal and matched at all ports

$$[S] = \begin{bmatrix} 0 & S_{12} & S_{13} \\ S_{21} & 0 & S_{23} \\ S_{31} & S_{32} & 0 \end{bmatrix}$$

For lossless network, must satisfy unitary condition

$$|S_{12}|^2 + |S_{13}|^2 = 1 \qquad S_{31}^* S_{32} = 0$$

$$|S_{21}|^2 + |S_{23}|^2 = 1 \qquad S_{21}^* S_{23} = 0$$

$$|S_{31}|^2 + |S_{32}|^2 = 1 \qquad S_{12}^* S_{13} = 0$$

The above equations must satisfy the following either

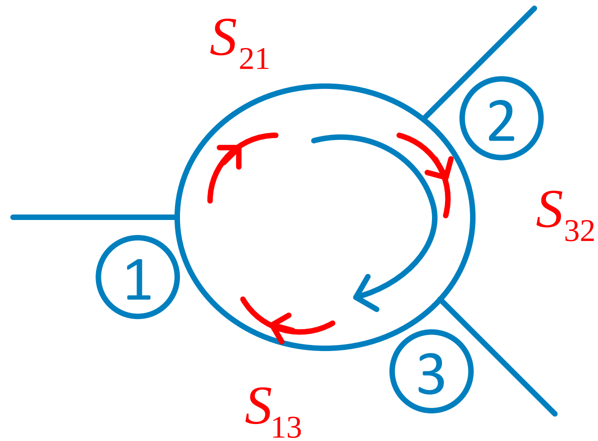
$$S_{12} = S_{23} = S_{31} = 0 \qquad |S_{21}| = |S_{32}| = |S_{13}| = 1$$

or

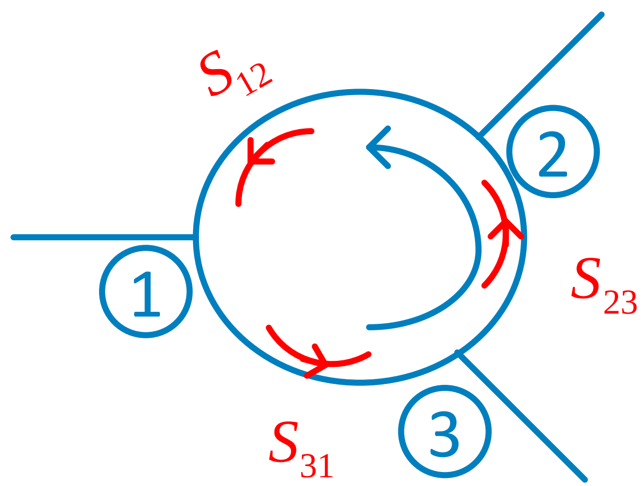
$$S_{21} = S_{32} = S_{13} = 0 \qquad |S_{12}| = |S_{23}| = |S_{31}| = 1$$

# Circulator (Nonreciprocal Network)

$$[S]=\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$




$$[S]=\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$




We can demonstrate here that any matched lossless three-port network must be nonreciprocal and, thus, a circulator.

# Another Alternative For Reciprocal Network

Only two ports are matched , then for lossless and reciprocal network

$$[S]=\begin{bmatrix} 0 & S_{12} & S_{13} \\ S_{12} & 0 & S_{23} \\ S_{13} & S_{23} & S_{33} \end{bmatrix}$$

For lossless network, must satisfy unitary condition

$$\left.\begin{aligned} |S_{12}|^2+|S_{13}|^2&=1 \\ |S_{12}|^2+|S_{23}|^2&=1 \end{aligned}\right\}$$

$$|S_{13}|^2+|S_{23}|^2+|S_{33}|^2=1$$

$$S_{13}^*S_{23}=0$$

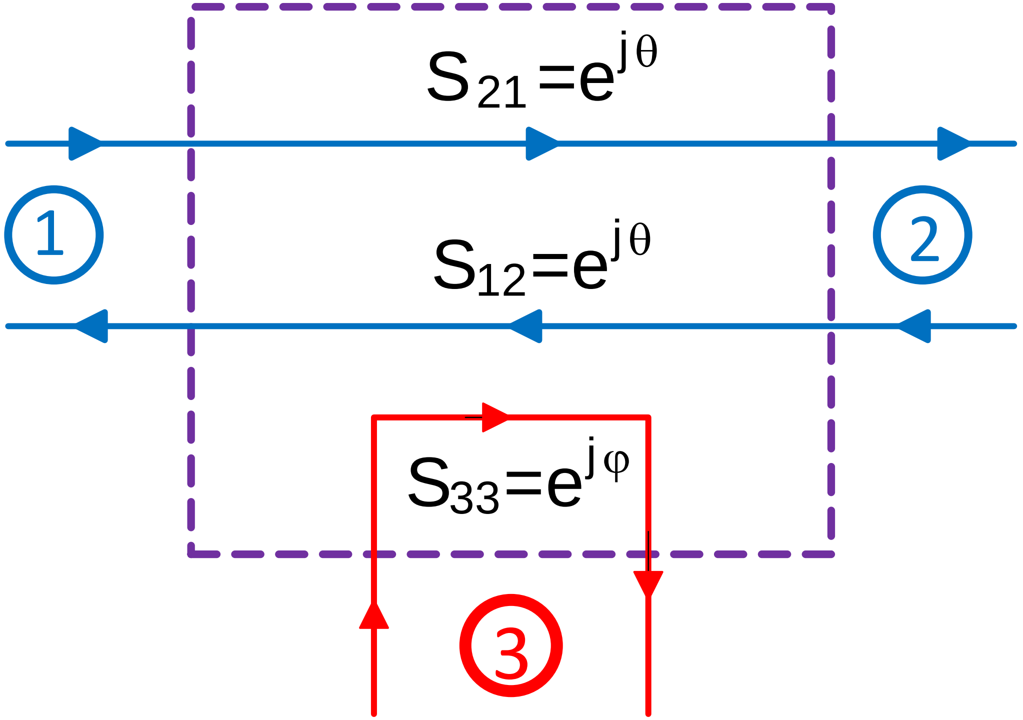
$$S_{12}^*S_{13}+S_{23}^*S_{33}=0$$

$$S_{23}^*S_{12}+S_{33}^*S_{13}=0$$

The two equations show that $|S_{13}|=|S_{23}|$
thus $S_{13}=S_{23}=0$
and $|S_{12}|=|S_{33}|=1$
These have satisfied all

# RECIPROCAL LOSSLESS NETWORK OF TWO MATCHED

$$[S] = \begin{bmatrix} 0 & e^{j\theta} & 0 \\ e^{j\theta} & 0 & 0 \\ 0 & 0 & e^{j\phi} \end{bmatrix}$$




Two port network with third port are isolated

# RECIPROCAL LOSSY NETWORK OF ALL PORTS MATCHED

- Finally, if the three-port network is allowed to be lossy, it can be reciprocal and matched at all ports;

- This is the case of the *resistive divider*

- In addition, a lossy three-port network can be made to have isolation between its output ports (e.g., $S_{23} = S_{32} = 0$).

# Three Port Networks (cont.)

1) Not possible → Lossless, Reciprocal, All ports matched

2) Circulator → Lossless, Non Reciprocal, All ports matched

3) Lossless power divider → Lossless, Reciprocal, One port matched

# Three Port Networks (cont.)

4) Lossy power divider → Lossy | Reciprocal | All ports matched

5) Two port network with third port are isolated → Lossless | Reciprocal | Two ports matched

# FOUR PORT NETWORK

Generally

$$[S]=\begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{bmatrix}$$

For reciprocal and lossless network

$$\boxed{\sum_{k=1}^{N} S_{ki} S_{ki}^{*} = 1}$$

$$\boxed{\sum_{k=1}^{N} S_{ki} S_{kj}^{*} = 0 \qquad for\ i \neq j}$$

$$|S_{11}|^2 + |S_{12}|^2 + |S_{13}|^2 + |S_{14}|^2 = 1$$

$$|S_{21}|^2 + |S_{22}|^2 + |S_{23}|^2 + |S_{24}|^2 = 1$$

$$|S_{31}|^2 + |S_{32}|^2 + |S_{33}|^2 + |S_{34}|^2 = 1$$

$$|S_{41}|^2 + |S_{42}|^2 + |S_{43}|^2 + |S_{44}|^2 = 1$$

R 1x R 2 $S_{11}S_{21}^{*} + S_{12}S_{22}^{*} + S_{13}S_{23}^{*} + S_{14}S_{24}^{*} = 0$

R1x R3 $S_{11}S_{31}^{*} + S_{12}S_{32}^{*} + S_{13}S_{33}^{*} + S_{14}S_{34}^{*} = 0$

R1x R4 $S_{11}S_{41}^{*} + S_{12}S_{42}^{*} + S_{13}S_{43}^{*} + S_{14}S_{44}^{*} = 0$

R 2x R3 $S_{21}S_{31}^{*} + S_{22}S_{32}^{*} + S_{23}S_{33}^{*} + S_{24}S_{34}^{*} = 0$

R2x R4 $S_{21}S_{41}^{*} + S_{22}S_{42}^{*} + S_{23}S_{43}^{*} + S_{24}S_{44}^{*} = 0$

R3x R4 $S_{31}S_{41}^{*} + S_{32}S_{42}^{*} + S_{33}S_{43}^{*} + S_{34}S_{44}^{*} = 0$

# MATCHED FOUR PORT NETWORK

Say all ports are matched and symmetrical network, then

$$[S]=\begin{bmatrix} 0 & S_{12} & S_{13} & S_{14} \\ S_{12} & 0 & S_{23} & S_{24} \\ S_{13} & S_{23} & 0 & S_{34} \\ S_{14} & S_{24} & S_{34} & 0 \end{bmatrix}$$

The unitarity condition become

$$|S_{12}|^2+|S_{13}|^2+|S_{14}|^2=1$$

$$|S_{12}|^2+|S_{23}|^2+|S_{24}|^2=1$$

$$|S_{13}|^2+|S_{23}|^2+|S_{34}|^2=1$$

$$|S_{14}|^2+|S_{24}|^2+|S_{34}|^2=1$$

R 1x R 2 $\quad S_{13}S_{23}^*+S_{14}S_{24}^*=0 \quad$ 1

R1x R3 $\quad S_{12}S_{23}^*+S_{14}S_{34}^*=0 \quad$ 2

R1x R4 $\quad S_{12}S_{24}^*+S_{13}S_{34}^*=0 \quad$ 3

R 2x R3 $\quad S_{12}S_{13}^*+S_{24}S_{34}^*=0 \quad$ 4

R2x R4 $\quad S_{12}S_{14}^*+S_{23}S_{34}^*=0 \quad$ 5

R3x R4 $\quad S_{13}S_{14}^*+S_{23}S_{24}^*=0 \quad$ 6

# TO CHECK VALIDITY

Multiply eq. 1 by $S_{24}^{*}$ and eq. 6 by $S_{13}^{*}$ , and substract to obtain

$$S_{14}^{*}\left(\left|S_{13}\right|^{2}-\left|S_{14}\right|^{2}\right)=0 \qquad 7$$

Multiply eq. 5 by $S_{34}$ and eq. 4 by $S_{13}$ , and substract to obtain

$$S_{23}\left(\left|S_{12}\right|^{2}-\left|S_{34}\right|^{2}\right)=0 \qquad 8$$

- Both equations 7 and 8 will be satisfy if $S_{14} = S_{23} = 0$ .
- This means that no coupling between port 1 and 4 , and between port 2 and 3 as happening in most directional couplers.

# ABCD TO S-PARAMETERS

$$S_{11} = \frac{A + B/Z_o - Z_o C - D}{A + B/Z_o + Z_o C + D}$$

$$S_{12} = \frac{2(AD - BC)}{A + B/Z_o + Z_o C + D}$$

$$S_{21} = \frac{2}{A + B/Z_o + Z_o C + D}$$

$$S_{22} = \frac{-A + B/Z_o - Z_o C + D}{A + B/Z_o + Z_o C + D}$$

For Symmetrical Network

$$S_{11} = S_{22} \qquad (A = D)$$

For Reciprocal Network

$$S_{12} = S_{21}$$

$$AD - BC = 1$$

# Directional Coupler

If all ports matched , symmetry and $S_{14}=S_{23}=0$ to be satisfied

$$[S]=\begin{bmatrix} 0 & S_{12} & S_{13} & 0 \\ S_{12} & 0 & 0 & S_{24} \\ S_{13} & 0 & 0 & S_{34} \\ 0 & S_{24} & S_{34} & 0 \end{bmatrix}$$

The equations reduce to 6 equations

$$S_{12}S_{24}^{*}+S_{13}S_{34}^{*}=0$$

$$S_{12}S_{13}^{*}+S_{24}S_{34}^{*}=0$$

i $|S_{12}|^2+|S_{13}|^2=1$

ii $|S_{12}|^2+|S_{24}|^2=1$

iii $|S_{13}|^2+|S_{34}|^2=1$

iv $|S_{24}|^2+|S_{34}|^2=1$

$\boxed{\text{By comparing equations i and ii yield } |S_{13}|=|S_{24}|}$

$\boxed{\text{By comparing equations ii and iii yield } |S_{12}|=|S_{34}|}$

# Directional Coupler (Contd.)

Simplified by choosing $S_{12} = S_{34} = \alpha$ ; $S_{13} = \beta e^{j\theta}$ and $S_{24} = \beta e^{j\varphi}$

R 2x R3 $S_{12}S_{13}^{*} + S_{24}S_{34}^{*} = 0$

Where $\theta + \varphi = \pi \pm 2n\pi$

$$[s] = \begin{bmatrix} 0 & \alpha & j\beta & 0 \\ \alpha & 0 & 0 & j\beta \\ j\beta & 0 & 0 & \alpha \\ 0 & j\beta & \alpha & 0 \end{bmatrix}$$

2 cases

1. Symmetry Coupler **$\theta = \varphi = \pi/2$**

2. Antisymmetry Coupler

**$\theta = 0, \varphi = \pi$**

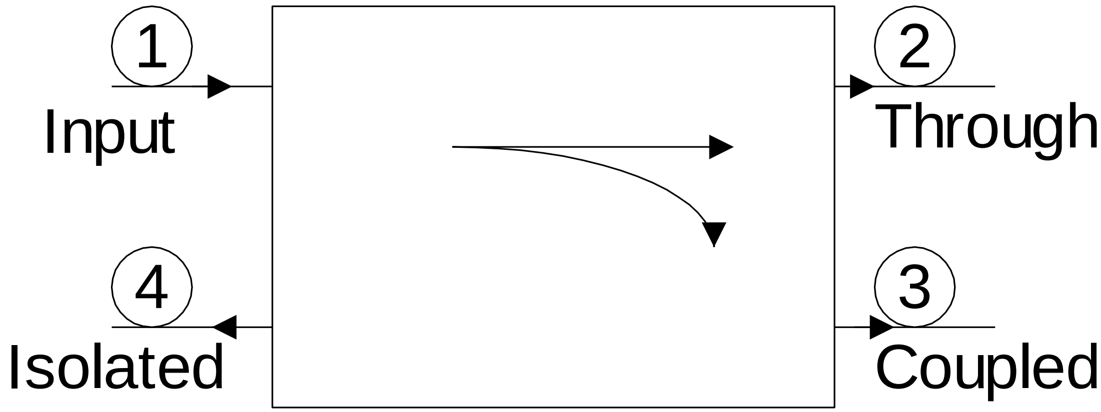
$$[s] = \begin{bmatrix} 0 & \alpha & \beta & 0 \\ \alpha & 0 & 0 & -\beta \\ \beta & 0 & 0 & \alpha \\ 0 & -\beta & \alpha & 0 \end{bmatrix}$$

Both satisfy

$$\alpha^2 + \beta^2 = 1$$

# PHYSICAL INTERPRETATION

$|S_{12}|^2$ = power deliver to port 2= $\alpha^2 = 1 - \beta^2$




$|S_{13}|^2$
= coupling factor
= $\beta^2$

Characterization of coupler

***Coupling=*** $C = 10\ log\dfrac{P_1}{P_3} = -20\log\beta \quad dB$

***Directivity=*** $D = 10\ log\dfrac{P_3}{P_4} = -20\log\dfrac{\beta}{|S_{14}|} \quad dB$

***Isolation =*** $I = 10\ log\dfrac{P_1}{P_4} = -20\log|S_{14}| \quad dB$

I = D + C dB

For ideal case
$|S_{14}|=0$

# PRACTICAL COUPLER

Hybrid 3 dB couplers

$\boxed{\theta = \varphi = \pi/2}$

$$[s] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & j & 0 \\ 1 & 0 & 0 & j \\ j & 0 & 0 & 1 \\ 0 & j & 1 & 0 \end{bmatrix}$$

$\alpha = \beta = 1/\sqrt{2}$

Magic -T and Rat-race couplers

$\boxed{\theta = 0, \varphi = \pi}$

$$[s] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \end{bmatrix}$$

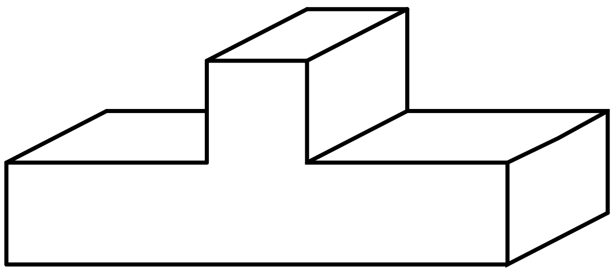
$\alpha = \beta = 1/\sqrt{2}$

# T-JUNCTION POWER DIVIDER

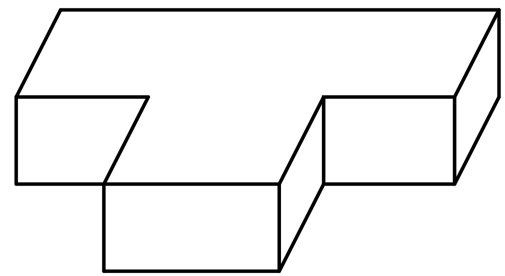
- ❖ The T-junction power divider is a simple three-port network that can be used for power division or power combining.

- ❖ It can be implemented in virtually any type of transmission line medium.

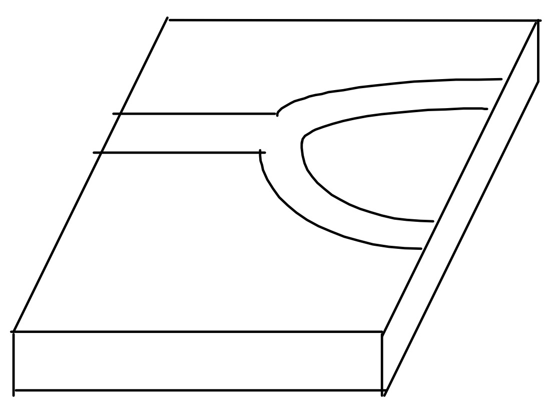
# T-JUNCTION POWER DIVIDER



E-plane T

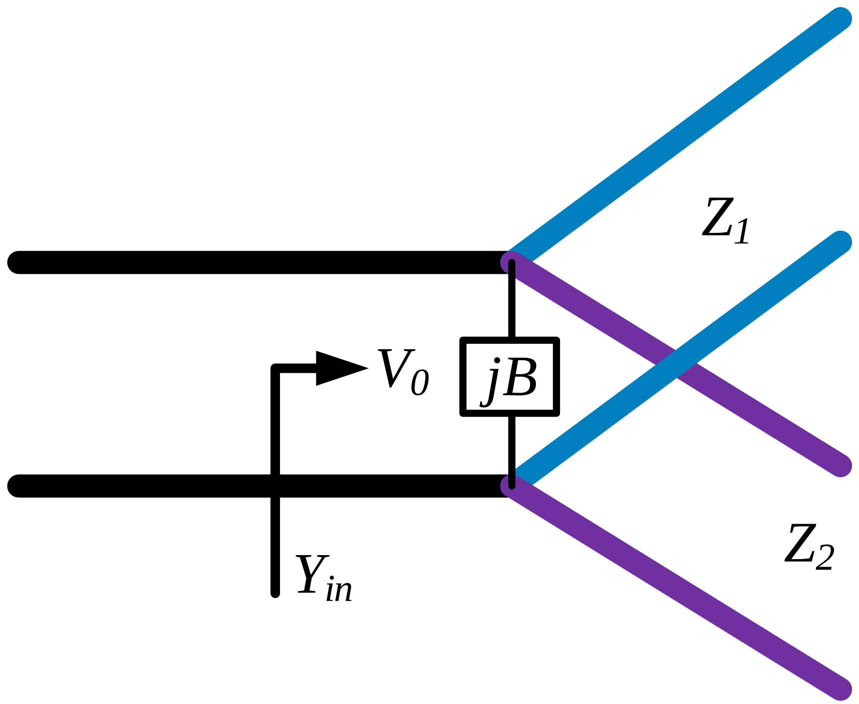


H-plane T



Microstrip T

# LOSSLESS DIVIDER




The lossless T-junctions of Figure can all be modeled as a junction of three transmission lines, as shown in Figure.

# T-MODEL

- The fringing fields and higher order modes associated with the discontinuity at such a junction, leading to stored energy that can be accounted for by a lumped susceptance, *B.*

- In order for the divider to be matched to the input line of characteristic impedance $Z_0$, we must have

Lossy line $$Y_{in} = jB + \frac{1}{Z_1} + \frac{1}{Z_2} = \frac{1}{Z_0}$$

Lossless line $$Y_{in} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

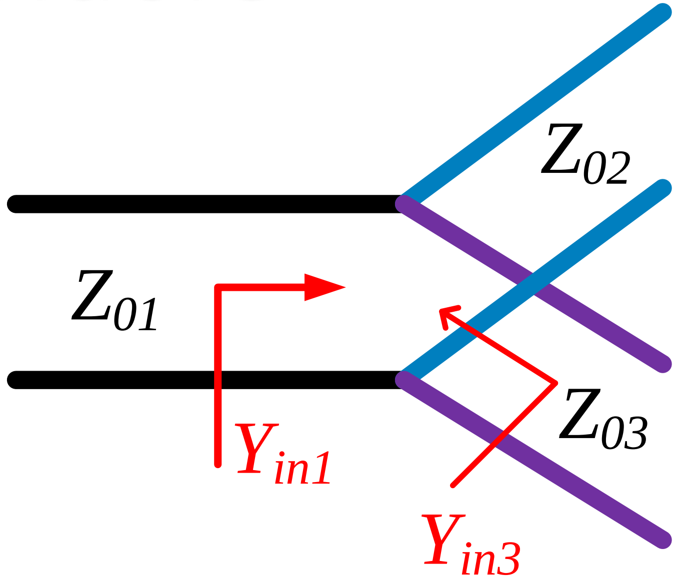
- In practice, if *B* is not negligible, some type of reactive tuning element can usually be added to the divider to cancel this susceptance, at least over a narrow frequency range.

If $Z_o = 50$,then for equally divided power, $Z_1 = Z_2=100$

# Power Dividers

## T-Junction: A lossless divider

$$Y_{\text{in}_1} = \frac{1}{Z_{02}} + \frac{1}{Z_{03}}$$




To match: $$Z_{01} = Z_{\text{in}_1} = \frac{1}{Y_{\text{in}_1}} = Z_{02} \,\|\, Z_{03} = \frac{Z_{02}Z_{03}}{Z_{02}+Z_{03}}$$

We then have: $$Y_{\text{in}_3} = \frac{1}{Z_{01}} + \frac{1}{Z_{02}} = \frac{Z_{02}+Z_{03}}{Z_{02}Z_{03}} + \frac{1}{Z_{02}} = \frac{Z_{02}+2Z_{03}}{Z_{02}Z_{03}} = \frac{1}{Z_{03}}\left(\frac{Z_{02}+2Z_{03}}{Z_{02}}\right)$$

Thus, $Y_{\text{in}_3} \neq \dfrac{1}{Z_{03}}$ Similarly, $Y_{\text{in}_2} \neq \dfrac{1}{Z_{02}}$

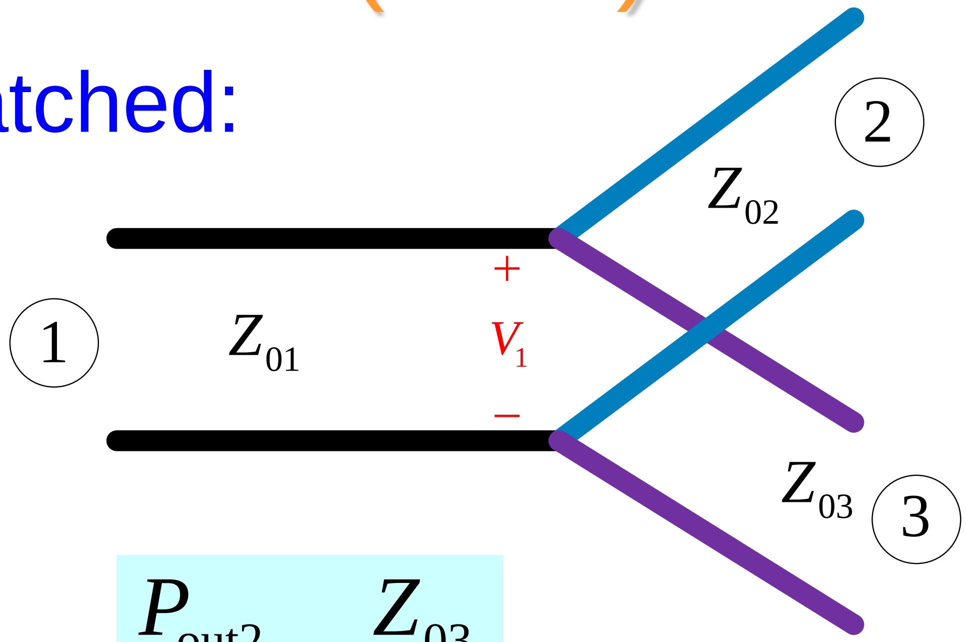
**If we match at port 1, we cannot match at the other ports!**

**Note:** Matching at ports 2 and 3 would be helpful when output lines 2 and 3 are not matched to their loads.

# Power Dividers (cont.)

## Assuming port 1 matched:

$$Z_{01} = \frac{Z_{02} Z_{03}}{Z_{02} + Z_{03}}$$




$$P_{\text{in}_1} = \frac{1}{2} \frac{|V_1|^2}{Z_{01}}$$

$$P_{\text{out}_2} = \frac{1}{2} \frac{|V_1|^2}{Z_{02}} = \frac{Z_{01}}{Z_{02}} P_{\text{in}_1}$$

$$\frac{P_{\text{out2}}}{P_{\text{out3}}} = \frac{Z_{03}}{Z_{02}}$$

$$P_{\text{out}_3} = \frac{1}{2} \frac{|V_1|^2}{Z_{03}} = \frac{Z_{01}}{Z_{03}} P_{\text{in}_1}$$

We can design the splitter to control the powers going into the two output lines.

For Equal power divider, $Z_{02} = Z_{03} = 2Z_{01}$

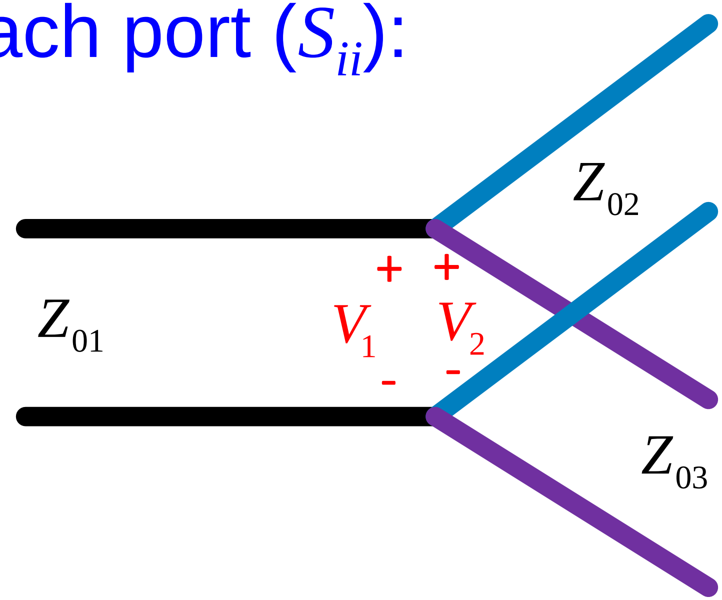
# Power Dividers (cont.)

## Examine the reflection at each port ($S_{ii}$):

$$S_{11} = \left.\frac{V_1^- / \sqrt{Z_{01}}}{V_1^+ / \sqrt{Z_{01}}}\right|_{a_2=a_3=0} = \left.\frac{V_1^-}{V_1^+}\right|_{a_2=a_3=0}$$

$$= \frac{Z_{in1} - Z_{01}}{Z_{in1} + Z_{01}} = \frac{Z_{02} \| Z_{03} - Z_{01}}{Z_{02} \| Z_{03} + Z_{01}}$$

$$= 0 \quad \left(\text{zero if port 1 is matched}\right)$$




$$S_{22} = \left.\frac{V_2^-}{V_2^+}\right|_{a_1=a_3=0}$$

$$= \frac{Z_{01} \| Z_{03} - Z_{02}}{Z_{01} \| Z_{03} + Z_{02}}$$

$$S_{33} = \left.\frac{V_3^-}{V_3^+}\right|_{a_1=a_2=0}$$

$$= \frac{Z_{01} \| Z_{02} - Z_{03}}{Z_{01} \| Z_{02} + Z_{03}}$$

**Note:** A match on port 1 requires:

$$Z_{01} < Z_{02}, \quad Z_{01} < Z_{03}$$

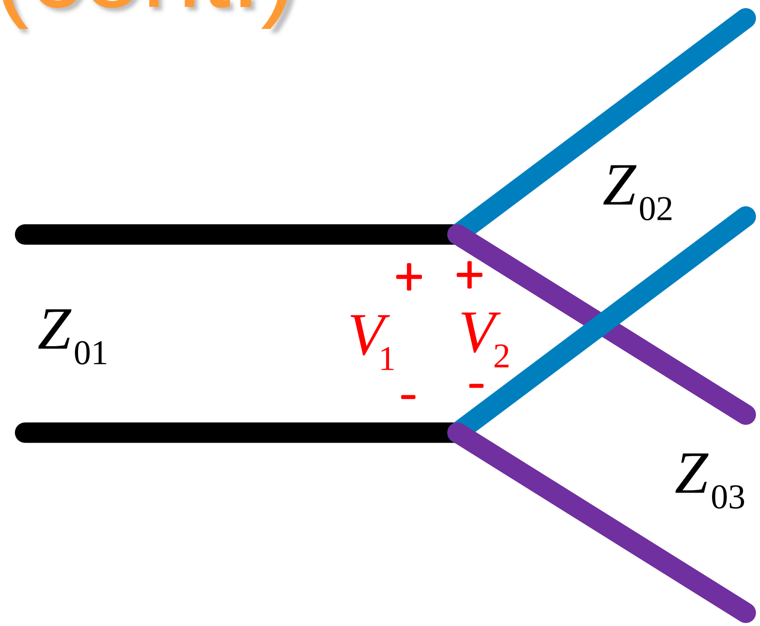
(The two output lines combine in parallel.)

The first term in the numerators must be less than the second term. $\Rightarrow S_{22} \neq 0, \quad S_{33} \neq 0$

# Power Dividers (cont.)

## Also, we have:

$$S_{21} = \left.\frac{\dfrac{V_2^-}{\sqrt{Z_{02}}}}{\dfrac{V_1^+}{\sqrt{Z_{01}}}}\right|_{a_2=a_3=0} = \frac{V_2^-}{V_1^+}\sqrt{\frac{Z_{01}}{Z_{02}}}$$




$$V_2^- / V_1^+ = V_2 / V_1^+ = V_1 / V_1^+$$

Also

$$V_1 = V_1^+\left(1+S_{11}\right) \Rightarrow V_1 / V_1^+ = \left(1+S_{11}\right)$$

Hence

$$S_{21} = \left(1+S_{11}\right)\sqrt{\frac{Z_{01}}{Z_{02}}} = S_{12}$$

Similarly:

$$S_{31} = \left(1+S_{11}\right)\sqrt{\frac{Z_{01}}{Z_{03}}} = S_{13}$$

$$S_{32} = S_{23} = \left(1+S_{22}\right)\sqrt{\frac{Z_{02}}{Z_{03}}}$$

# Power Dividers (cont.)

If port 1 is matched: $Z_{01} = \dfrac{Z_{02}Z_{03}}{Z_{02}+Z_{03}}$

$$\Rightarrow S_{11} = 0 \ ; \quad S_{22} = \frac{Z_{01}\|Z_{03} - Z_{02}}{Z_{01}\|Z_{03} + Z_{02}} \ ; \quad S_{33} = \frac{Z_{01}\|Z_{02} - Z_{03}}{Z_{01}\|Z_{02} + Z_{03}}$$

**The output ports 2 and 3 are <u>not</u> Matched.**

$$S_{21} = S_{12} = (1+S_{11})\sqrt{\frac{Z_{01}}{Z_{02}}} = \sqrt{\frac{Z_{01}}{Z_{02}}} = \sqrt{\frac{Z_{03}}{Z_{02}+Z_{03}}}$$

$$S_{31} = S_{13} = (1+S_{11})\sqrt{\frac{Z_{01}}{Z_{03}}} = \sqrt{\frac{Z_{01}}{Z_{03}}} = \sqrt{\frac{Z_{02}}{Z_{02}+Z_{03}}}$$

$$[S] = \begin{pmatrix} 0 & S_{12} & S_{13} \\ S_{12} & S_{22} & S_{23} \\ S_{13} & S_{23} & S_{33} \end{pmatrix}$$

From last slide: $S_{32} = S_{23} = (1+S_{22})\sqrt{\dfrac{Z_{02}}{Z_{03}}}$

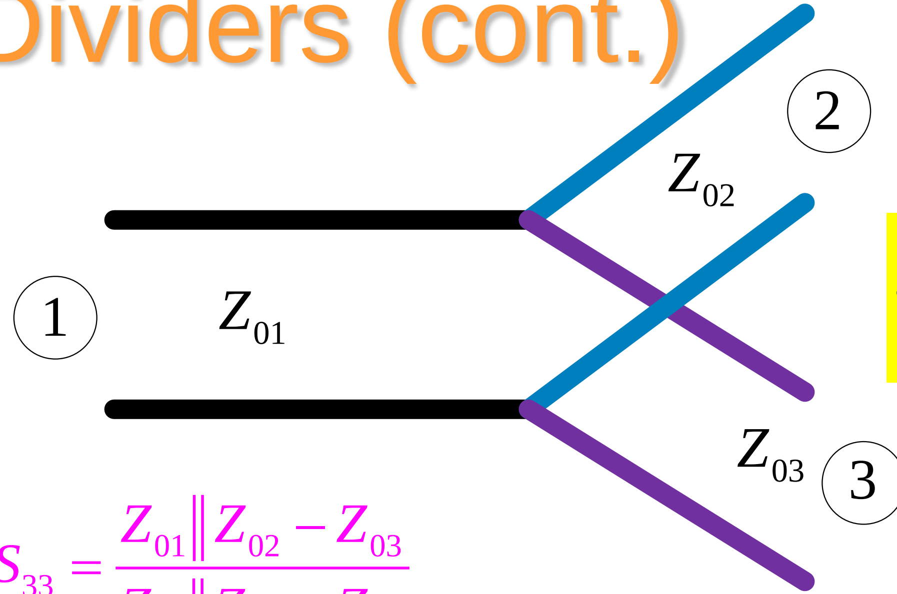
Only $S_{11} = 0$

The output ports 2 and 3 are <u>not</u> isolated.

# Power Dividers (cont.)

## Summary

$$Z_{01} = \frac{Z_{02} Z_{03}}{Z_{02} + Z_{03}}$$




$$\frac{P_{out3}}{P_{out2}} = \frac{Z_{02}}{Z_{03}}$$

$$S_{11} = 0 \;;\quad S_{22} = \frac{Z_{01} \| Z_{03} - Z_{02}}{Z_{01} \| Z_{03} + Z_{02}} \;;\quad S_{33} = \frac{Z_{01} \| Z_{02} - Z_{03}}{Z_{01} \| Z_{02} + Z_{03}}$$

$$S_{21} = S_{12} = \sqrt{\frac{Z_{03}}{Z_{02} + Z_{03}}}$$

$$S_{31} = S_{13} = \sqrt{\frac{Z_{02}}{Z_{02} + Z_{03}}}$$

$$S_{32} = S_{23} = (1 + S_{22}) \sqrt{\frac{Z_{02}}{Z_{03}}}$$

- ❖ The input port is matched, but not the output ports.

- ❖ The output ports are not isolated.
- ❖ Waves reflected from devices on ports 2 and 3 with cause <u>interference</u> with the other devices.

# Power Dividers (cont.)

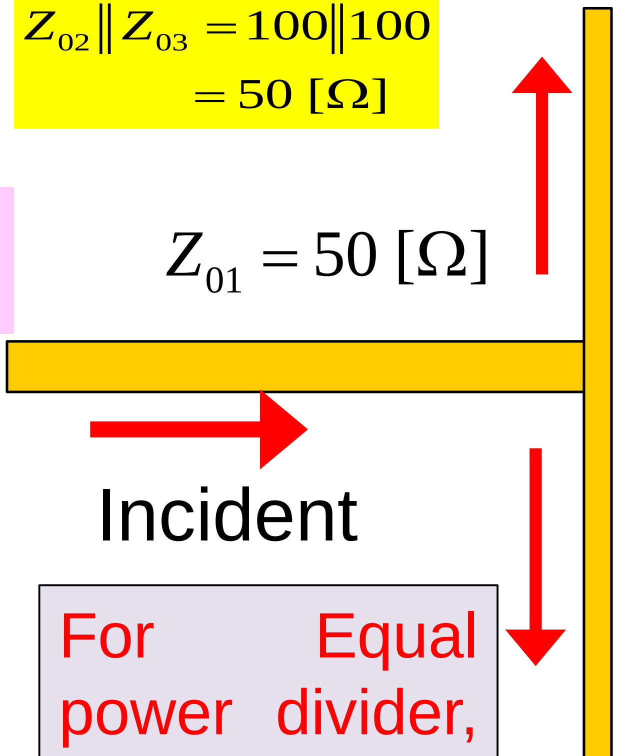
## Example: Microstrip T-junction Equal power divider

$$S_{11} = \frac{Z_{02} \| Z_{03} - Z_{01}}{Z_{02} \| Z_{03} + Z_{01}} = 0$$

$$Z_{02} \| Z_{03} = 100 \| 100 = 50\ [\Omega]$$

$$Z_{02} = 2Z_{01} = 100\ [\Omega]$$

$$S_{22} = S_{33} = \frac{Z_{01} \| Z_{02} - Z_{03}}{Z_{01} \| Z_{02} + Z_{03}} = -\frac{1}{2}$$




$$Z_{01} \| Z_{02} = Z_{01} \| Z_{03} = 50 \| 100 = 33.333\ [\Omega]$$

$$S_{21} = S_{12} = \sqrt{\frac{Z_{03}}{Z_{02} + Z_{03}}} = \sqrt{\frac{1}{2}}$$

$$S_{32} = S_{23} = (1 + S_{22})\sqrt{\frac{Z_{02}}{Z_{03}}} = \frac{1}{2}$$

$$S_{31} = S_{13} = \sqrt{\frac{Z_{02}}{Z_{02} + Z_{03}}} = \sqrt{\frac{1}{2}}$$

For Equal power divider, $Z_{02} = Z_{03} = 2Z_{01}$

$$Z_{03} = 2Z_{01} = 100\ [\Omega]$$

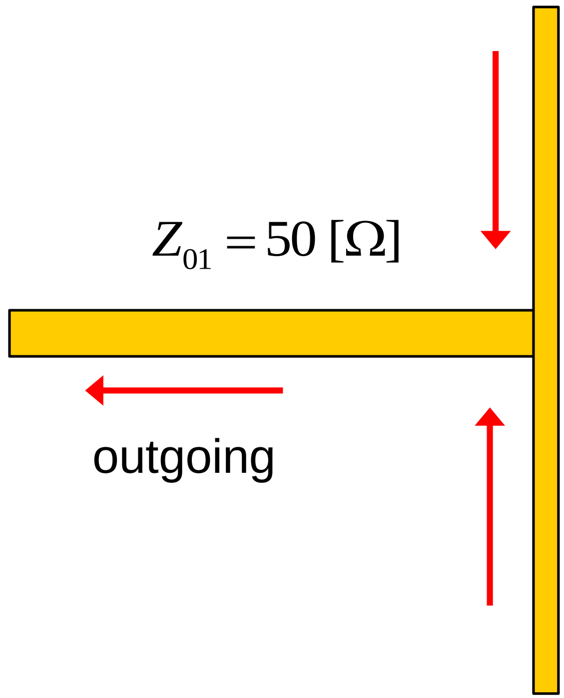
**Note:** Quarter-wave transformers could be put on the output lines to bring the final output lines back to 50 [Ω].

# Power Dividers (cont.)

The matched power divider also works as a matched power combiner.

$$Z_{02} = 100\ [\Omega]$$

$$Z_{01} = 50\ [\Omega]$$




$$Z_{01} \| Z_{02} = Z_{01} \| Z_{03}$$
$$= 50 \| 100 = 33.333\ [\Omega]$$

$$Z_{03} = 100\ [\Omega]$$

$$S_{11} = 0$$

$$S_{22} = S_{33} = -\frac{1}{2}$$

$$S_{21} = S_{12} = \sqrt{\frac{1}{2}}$$

$$S_{31} = S_{13} = \sqrt{\frac{1}{2}}$$

$$S_{32} = S_{23} = \frac{1}{2}$$

$$\begin{aligned} b_3 &= \cancel{a_1} S_{31} + a_2 S_{32} + a_3 S_{33} \\ &= a_3 \left( S_{33} + S_{32} \right) \\ &= a_3 \left( -\frac{1}{2} + \frac{1}{2} \right) \\ &= 0 \end{aligned}$$

Equal waves are incident from ports 2 and 3 ($a_2 = a_3$).

There is no reflection if <u>equal</u> waves are incident on ports 2 and 3, and port 1 is matched.

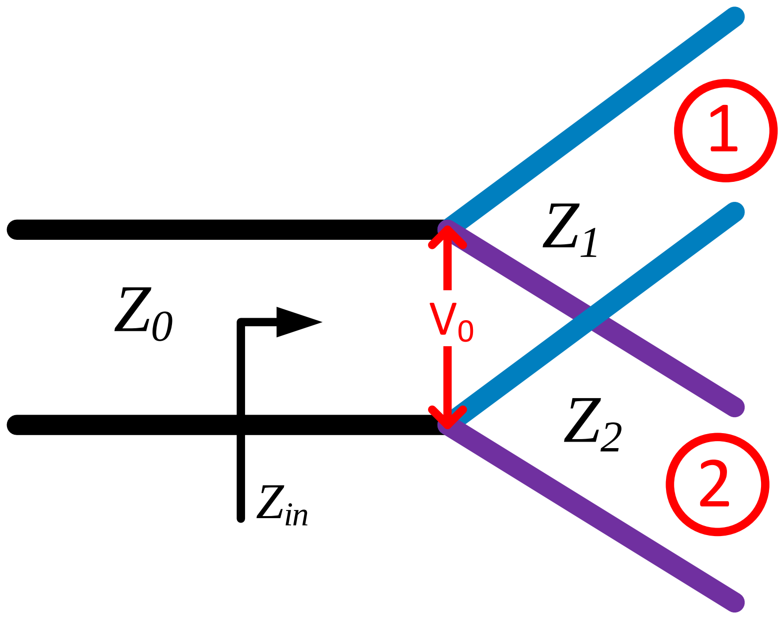
# EXAMPLE

If source impedance equal to 50 ohm and the power to be divided into 2:1 ratio. Determine $Z_1$ and $Z_2$

$$P_{in} = \frac{1}{2}\frac{V_o^2}{Z_o}$$

$$P_1 = \frac{1}{2}\frac{V_o^2}{Z_1} = \frac{1}{3}P_{in} = \frac{1}{2}\frac{1}{3}\frac{V_o^2}{Z_o}$$

$$Z_1 = 3Z_o = 150\Omega$$




$$P_2 = \frac{1}{2}\frac{V_o^2}{Z_2} = \frac{1}{2}\frac{2}{3}\frac{V_o^2}{Z_o}P_{in}$$

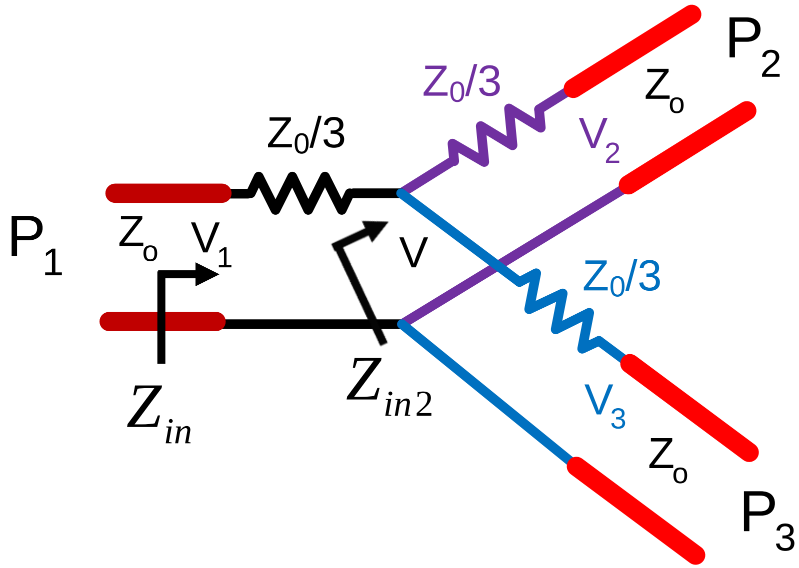
$$Z_2 = \frac{3Z_o}{2} = 75\Omega$$

$$Z_{in} = Z_1 // Z_2 = 50\Omega = Z_o$$

# RESISTIVE DIVIDER

- If a three-port divider contains lossy components, it can be made to be matched at all ports, although the two output ports may not be isolated.
- An equal-split (-3 dB) divider is shown in Figure, using lumped-element resistors, but unequal power division ratios are also possible.
- Assuming that all ports are terminated in the characteristic impedance $Z_0$.

# RESISTIVE DIVIDER




$$Z_{in2} = Z_o + \frac{Z_o}{3} = \frac{4Z_o}{3} = Z_{in3}$$

Then the input impedance of the divider is

$$Z_{in} = \frac{Z_o}{3} + \frac{4Z_o}{3} \bigg\| \frac{4Z_o}{3} = \frac{Z_o}{3} + \frac{2Z_o}{3} = Z_o$$

$$V = \frac{2Z_o/3}{Z_o/3 + 2Z_o/3} V_1 = \frac{2}{3} V_1$$

$$V_2 = V_3 = \frac{Z_o}{Z_o + Z_o/3} V = \frac{3}{4} V = \frac{1}{2} V_1$$

$$P_{in} = \frac{1}{2} \frac{V_1^2}{Z_o}$$

$$P_2 = P_3 = \frac{1}{2} \frac{(1/2V_1)^2}{Z_o} = \frac{1}{4} P_{in}$$

# RESISTIVE DIVIDER

$$Power\ \ loss = P_{Loss} = P_{in} - \left(P_2 + P_3\right) = P_{in} - \frac{1}{2}P_{in} = \frac{1}{2}P_{in}$$

- Thus, $S_{21}$= $S_{31}$= $S_{23}$= 1/2, which is -6 dB below the input power level.
- The network is reciprocal, so the scattering matrix is symmetric, and can be written as

$$[S] = \frac{1}{2}\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

# THE WILKINSON POWER DIVIDER

- ❖ The lossless T-junction divider suffers from the problem of not being matched at all port.

- ❖ The lossless T-junction divider does not have any isolation between output ports.

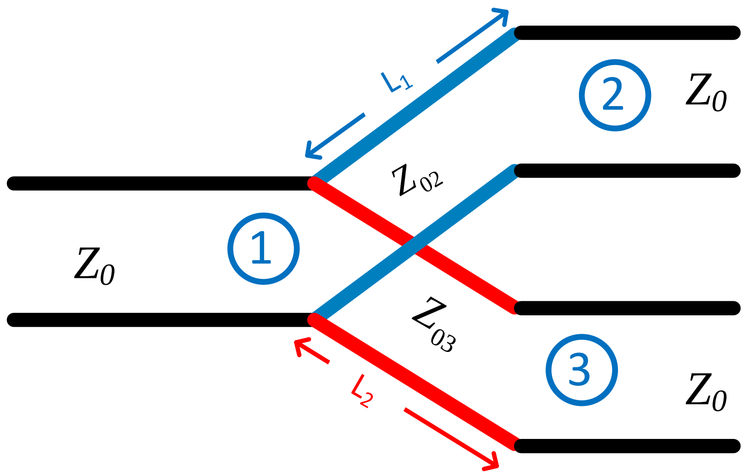
- ❖ The resistive divider can be matched at all ports, but even though it is not lossless, isolation is still not achieved.

# THE WILKINSON POWER DIVIDER

- We know that a lossy three-port network can be made having all ports matched with isolation between the output.
- The Wilkinson power divider is such a network, with the useful property of being lossless when the output ports are matched.
- In the Wilkinson power divider, only reflected power from two out ports is dissipated.

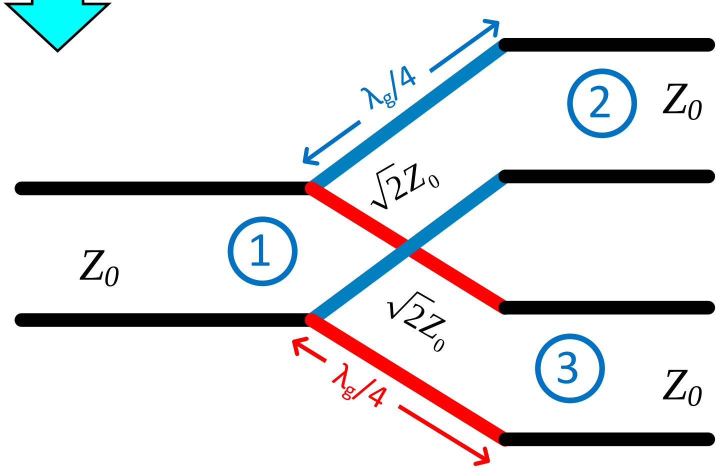
# THE WILKINSON POWER DIVIDER

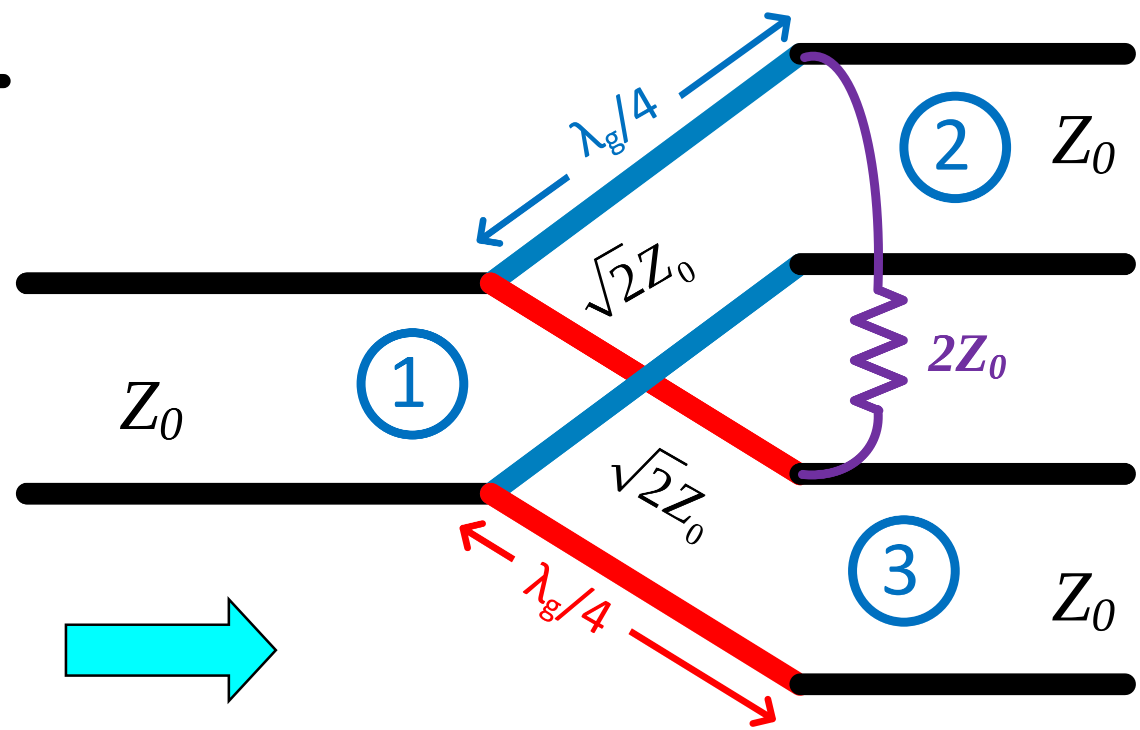
- Two output are not Matched
- Two output are not isolated




$$L_1 = L_2 = \lambda_g / 4$$

$$Z_{02} = Z_{03} = \sqrt{2}Z_0$$



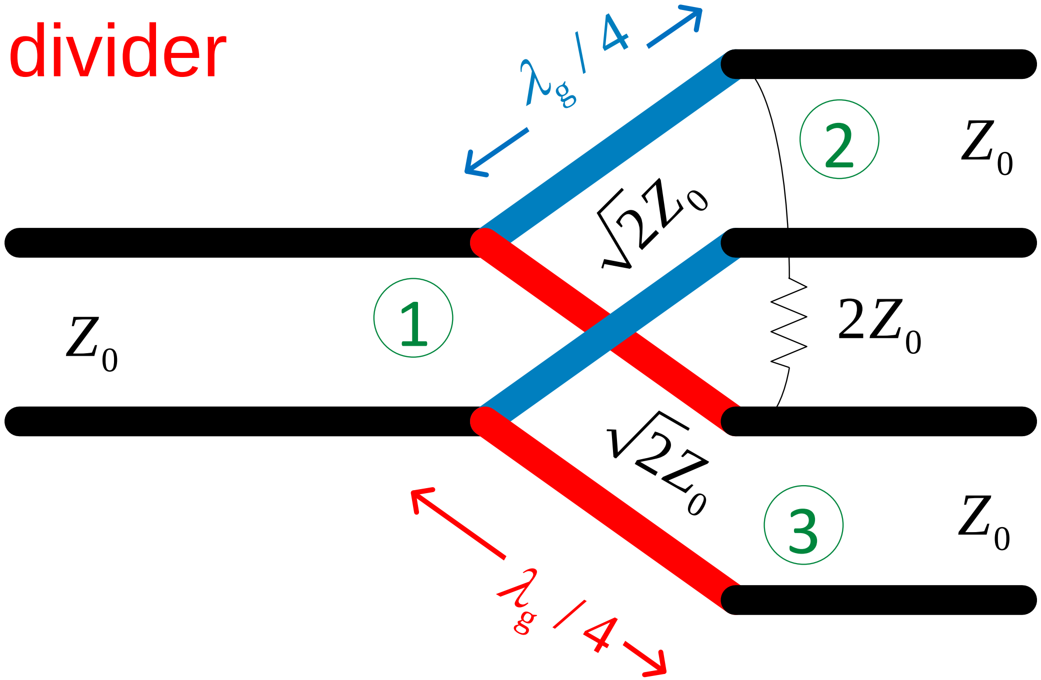




- For isolation between output, $2Z_0$ are added

# Wilkinson Power Divider

## Equal-split (3 dB) power divider

(The Wilkenson can also be designed to have an unequal split.)




- ❖ All ports matched ($S_{11} = S_{22} = S_{33} = 0$)
- ❖ Output ports are isolated ($S_{23} = S_{32} = 0$)

***Note:*** No power is lost in going from port 1 to ports 2 and 3:

$$[S] = \frac{-j}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

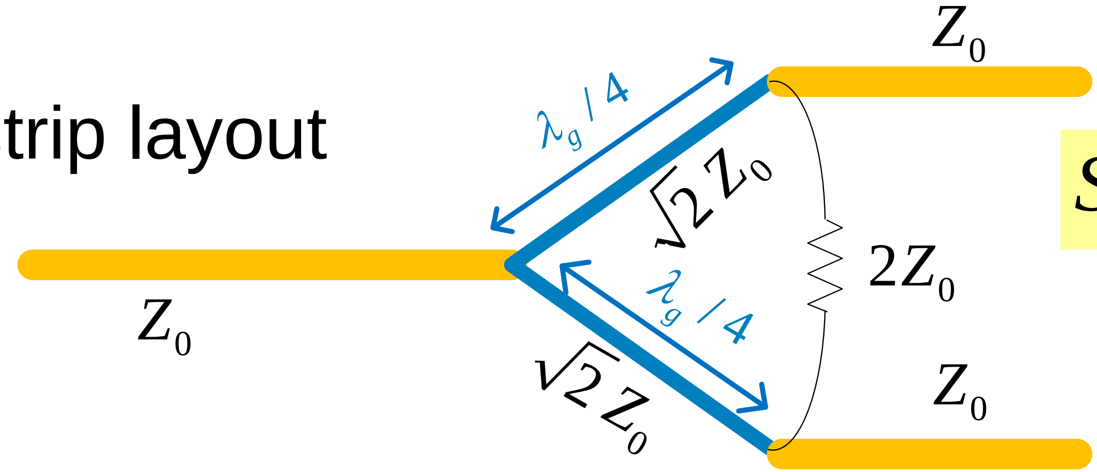
$$|S_{21}|^2 = |S_{31}|^2 = \frac{1}{2}$$

Obviously not unitary

# Wilkinson Power Divider (cont.)

$$[S] = \frac{-j}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Microstrip layout



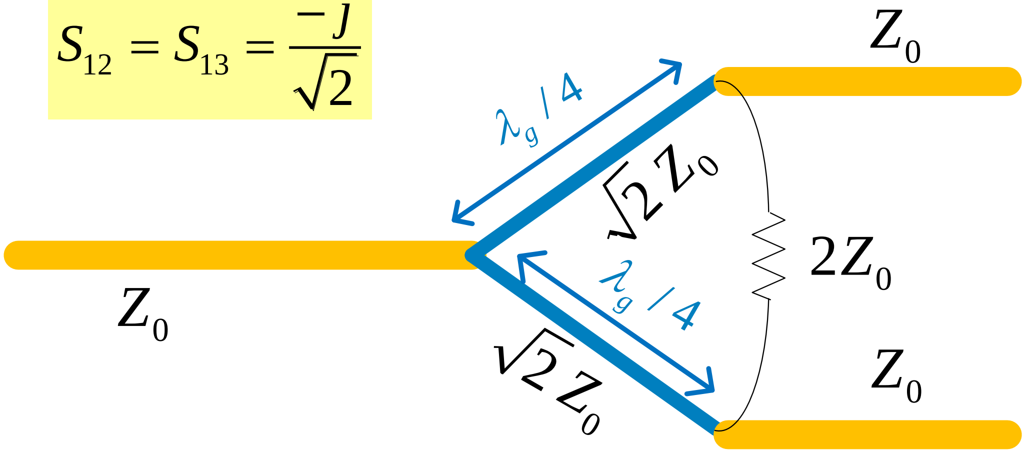

$$S_{11} = S_{22} = S_{33} = 0$$

$$S_{32} = S_{23} = 0$$

All three ports are matched, and the output ports are isolated.

# Wilkinson Power Divider (cont.)

$$S_{12} = S_{13} = \frac{-j}{\sqrt{2}}$$




$$[S] = \frac{-j}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

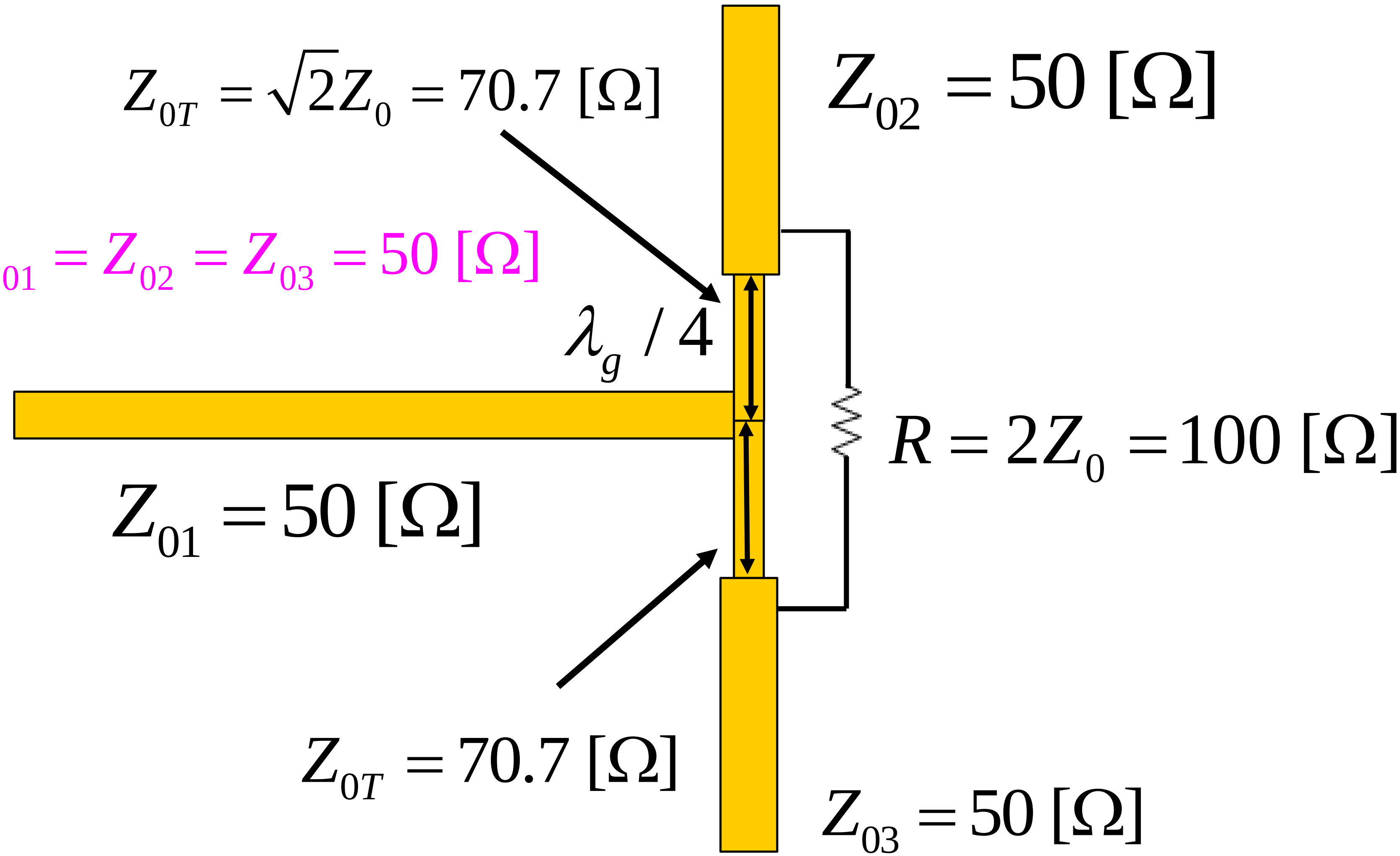
$$S_{21} = S_{31} = \frac{-j}{\sqrt{2}}$$

- When a wave is incident from port 1, half of the total incident power gets transmitted to each output port (no loss of power).
- When a wave is incident from port 2 or port 3, half of the power gets transmitted to port 1 and half gets absorbed by the resistor.
- But nothing gets through to the other output port (the two output ports are isolated from each other).

# Wilkinson Power Divider (cont.)

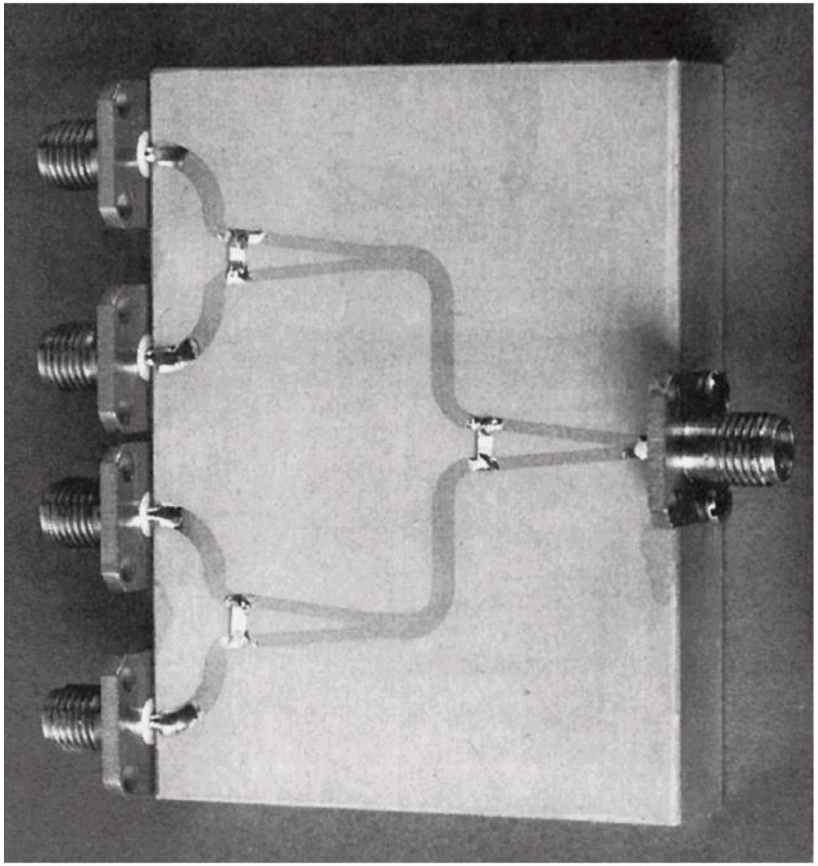
## Example: Microstrip Wilkinson power divider

$Z_0 = Z_{01} = Z_{02} = Z_{03} = 50\ [\Omega]$

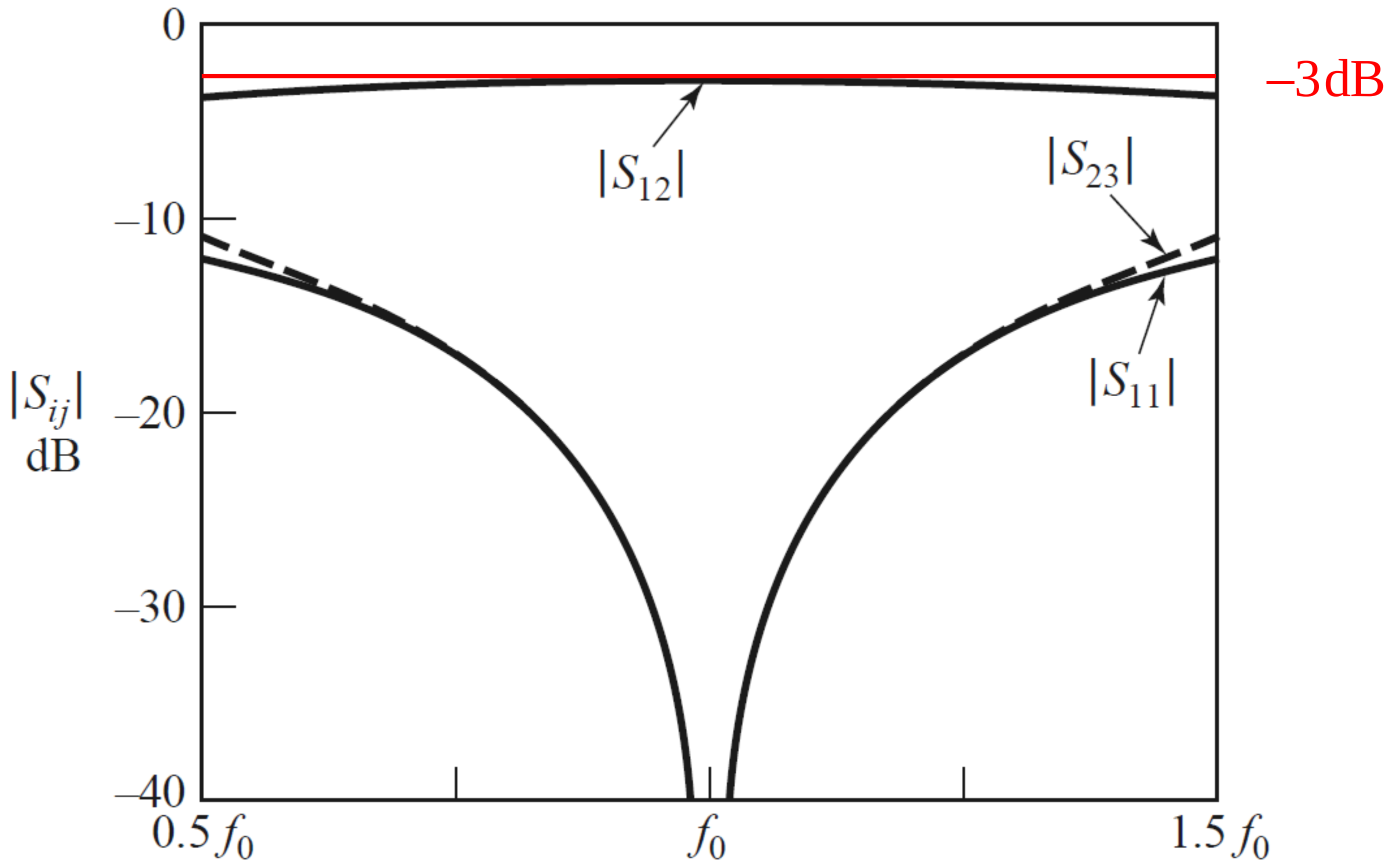


# Wilkinson Power Divider (cont.)

Figure 7.15 of Pozar
Photograph of a four-way corporate power divider network using three microstrip Wilkinson power dividers. Note the isolation chip resistors.



# Wilkinson Power Divider




- Figure depict Frequency response of an equal-split Wilkinson power divider.
- Port 1 is the input port; ports 2 and 3 are the output ports.

# ODD AND EVEN MODE ANALYSIS

**Usually use for analyzing a symmetrical four port network**

(1) Excitation:
- Equal ,in phase excitation →even mode
- Equal ,out of phase excitation → odd mode

(2) Draw intersection line for symmetry and apply
- short circuit for odd mode
- Open circuit for even mode

(3) Also can apply EM analysis of the structure
- Tangential E field zero – odd mode
- Tangential H field zero – even mode

(4) Single excitation at one port= even mode + odd mode

# Wilkinson Power Divider (cont.)

**The analysis of the Wilkinson power divider is given here.**

❖ Even/odd mode analysis is used to analyze the Wilkinson power divider.

❖ This requires two separate analyzes (even and odd).

❖ Each analysis involves only a <u>two-port</u> device instead of a three-port device.

# Wilkinson Power Divider (cont.)

- Even and odd analysis is used to analyze the structure when port 2 is excited.

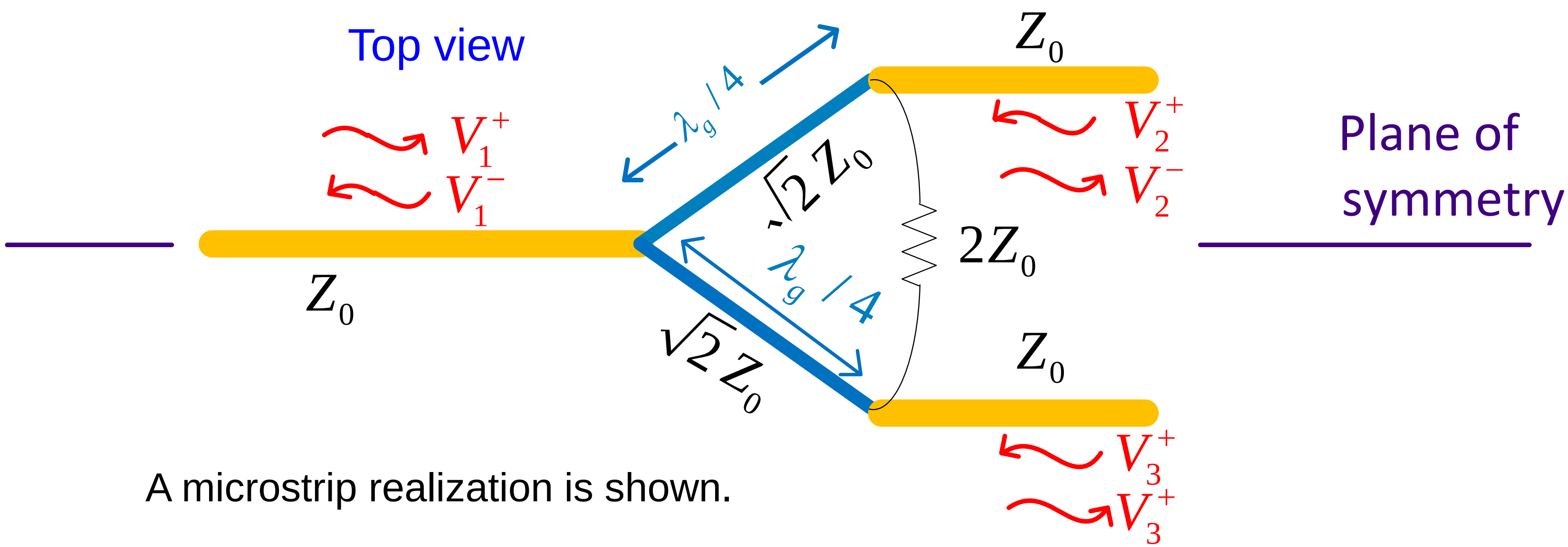
$$\Rightarrow \text{ To determine } S_{22},\ S_{32}$$

- Only even analysis is needed to analyze the structure when port 1 is excited.

$$\Rightarrow \text{ To determine } S_{11},\ S_{21}$$

- The other five components of the $S$ matrix can be found by using physical symmetry and reciprocity (the symmetry of the $S$ matrix).

# Wilkinson Power Divider (cont.)



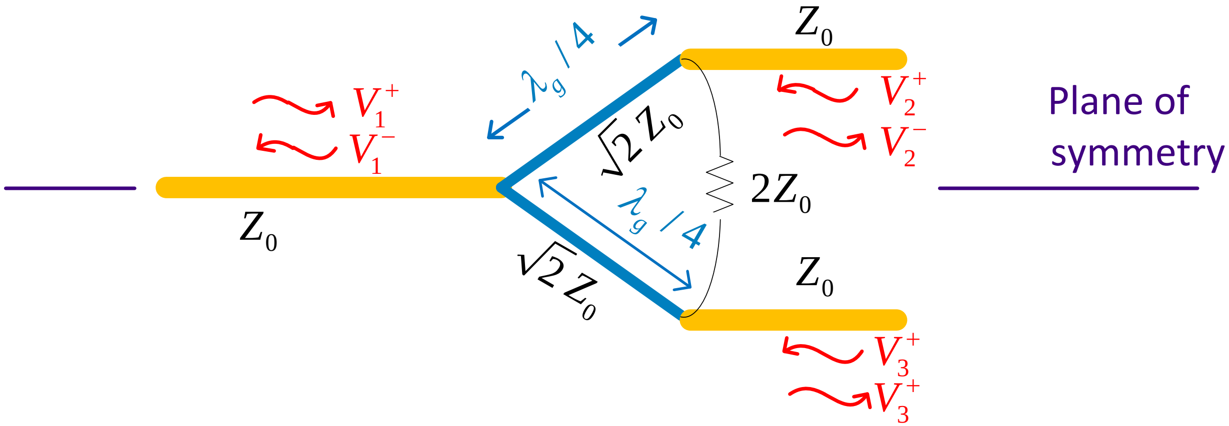

A microstrip realization is shown.

Split structure along plane of symmetry (POS)

Even $\Rightarrow$ voltage even about POS $\Rightarrow$ place OC along POS

Odd $\Rightarrow$ voltage odd about POS $\Rightarrow$ place SC along POS

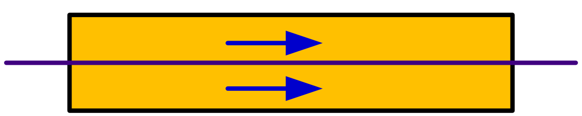
# Wilkinson Power Divider (cont.)




How do you split a transmission line? (This is needed for the even case.)

- Voltage is the same for each half of line ($V$)
- Current is halved for each half of line ($I/2$)

Top view




$Z_0$ microstrip line

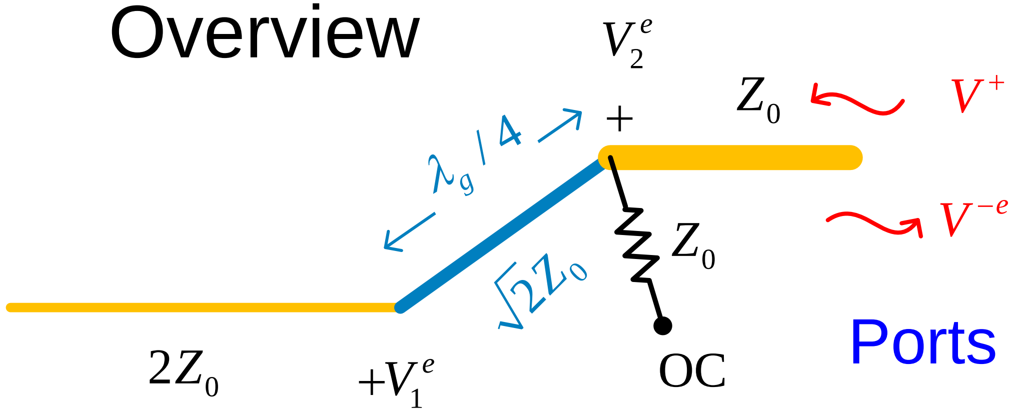
For each half

$$\Rightarrow Z_0^h = \frac{V}{I/2} = 2Z_0$$

# Wilkinson Power Divider (cont.)

Port 2 Excitation
"even" problem

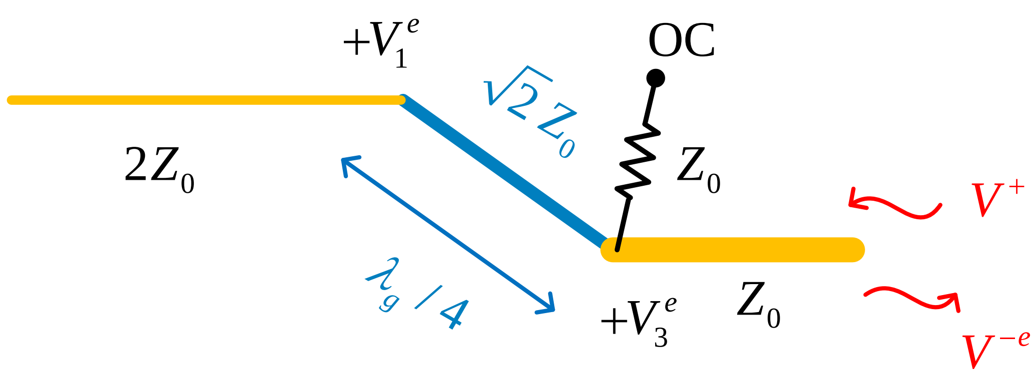
Overview




**Note**:
The $2Z_0$ resistor has been split into two $Z_0$ resistors in series.

Ports 2 and 3 are excited in phase.

$$\Rightarrow S_{22}^e,\ S_{32}^e$$




$$S_{22}^e \equiv \frac{V^{-e}}{V^+}$$

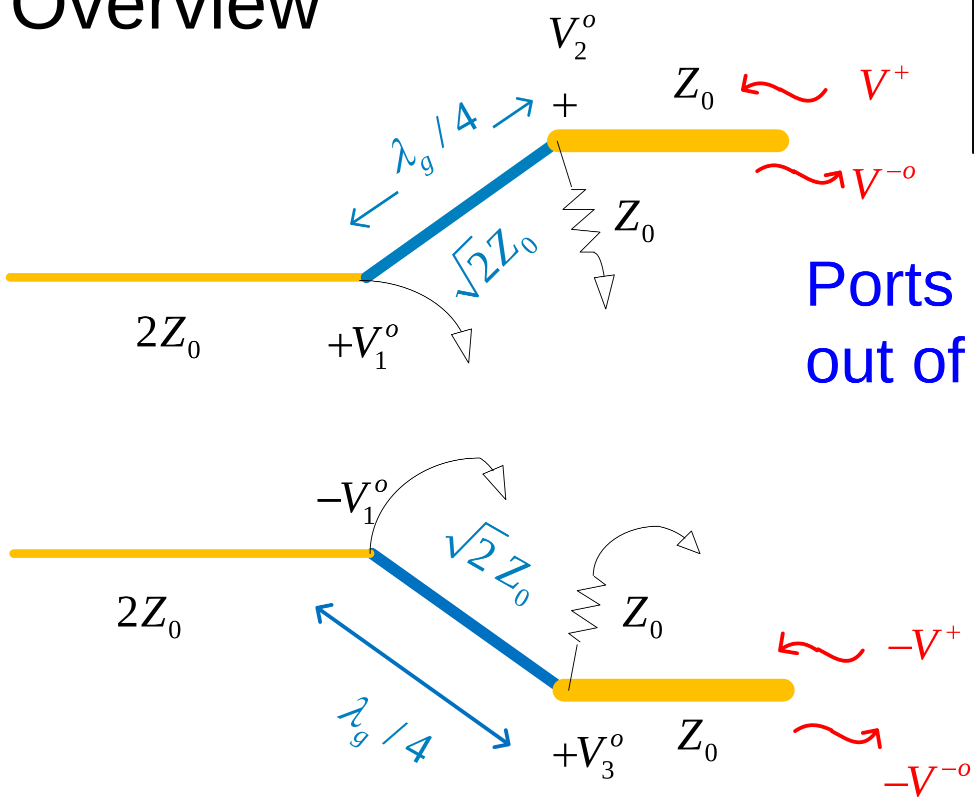
$$S_{32}^e \equiv \frac{V^{-e}}{V^+} = S_{22}^e$$

Note : $V_3^e = V_2^e$

# Wilkinson Power Divider (cont.)

**Note:**
The $2Z_0$ resistor has been split into two $Z_0$ resistors in series.

Port 2 Excitation
"odd" problem
Overview




Ports 2 and 3 are excited 180° out of phase.

$$\Rightarrow S_{22}^o,\ S_{32}^o$$

$$S_{22}^o \equiv \frac{V^{-o}}{V^+}$$

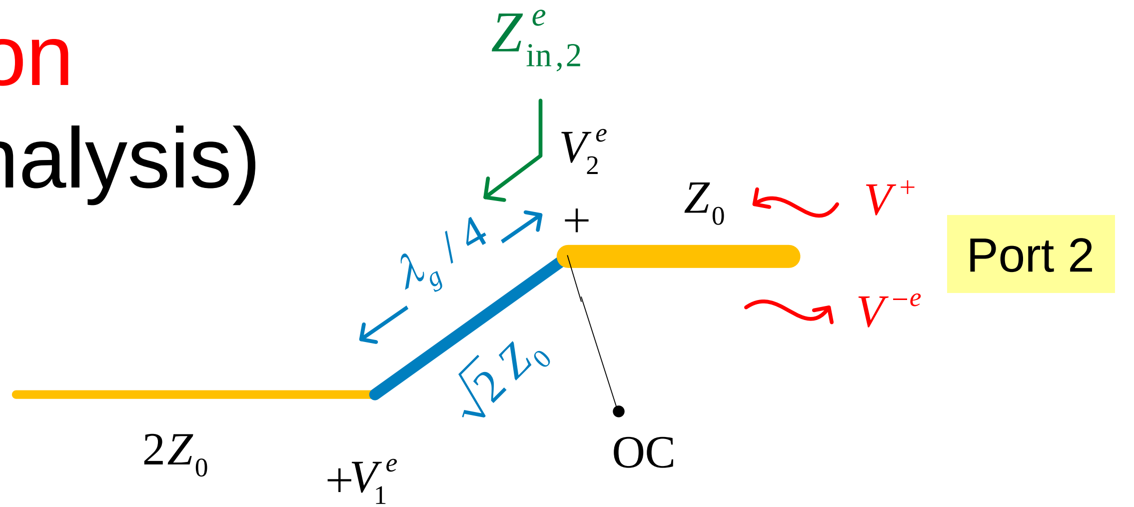
$$S_{32}^o \equiv \frac{-V^{-o}}{V^+} = -S_{22}^o$$

Note: $V_1^o = 0, \quad V_3^o = -V_2^o$

# Wilkinson Power Divider (cont.)

## Port 2 Excitation

## "even" problem (Analysis)




$$Z_{\text{in}\,2}^{e} = \frac{Z_T^2}{Z_L}$$

$$Z_{\text{in}\,2}^{e} = \frac{\left(\sqrt{2}\,Z_0\right)^2}{2Z_0} = Z_0$$

(quarter-wave transformer)

$$\Rightarrow S_{22}^{e} = \frac{Z_{\text{in}\,2}^{e} - Z_0}{Z_{\text{in}\,2}^{e} + Z_0} = 0$$

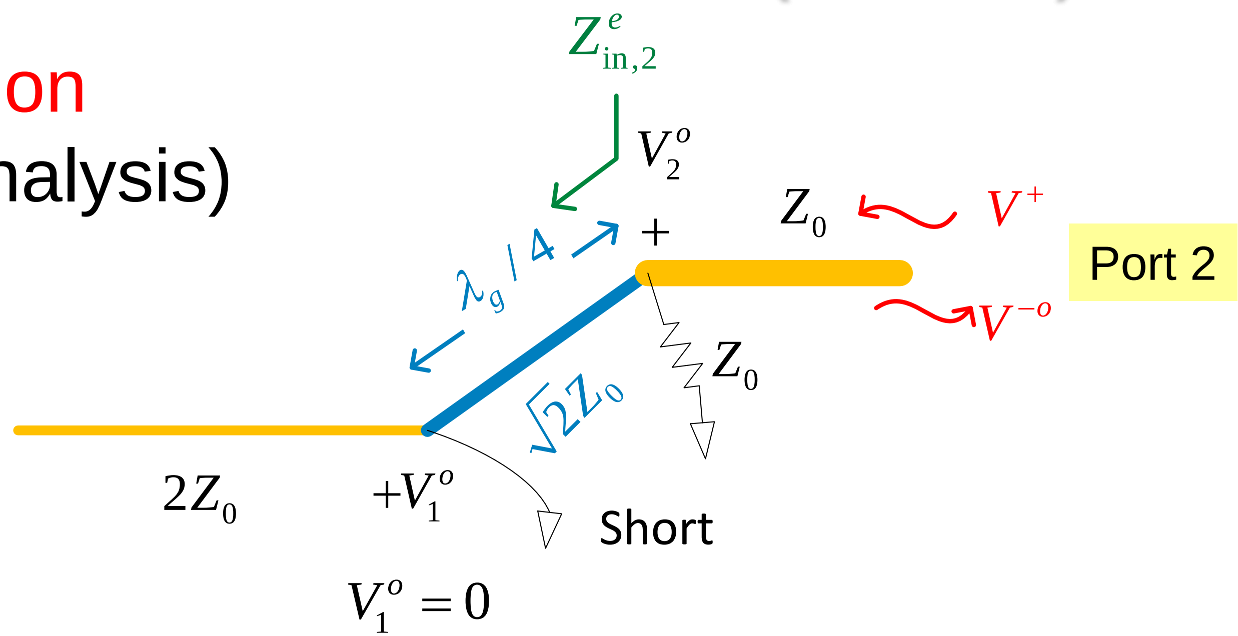
Also, by physical symmetry:

$$S_{33}^{e} = S_{22}^{e} = 0$$

Also, in the even case: $S_{32}^{e} = S_{22}^{e} \Rightarrow S_{32}^{e} = 0$

# Wilkinson Power Divider (cont.)

## Port 2 Excitation
## "odd" problem (Analysis)




$$Z_{\text{in}2}^{o} = \frac{Z_T^2}{Z_L}$$

$$= \frac{\left(\sqrt{2}\,Z_0\right)^2}{0} = \infty$$

$$Z_{\text{in}2}^{o} = \infty \,\|\, Z_0 = Z_0$$

$$\Rightarrow S_{22}^{o} = \frac{Z_{\text{in}2}^{o} - Z_0}{Z_{\text{in}2}^{o} + Z_0} = 0$$

Also, by physical symmetry:

$$S_{33}^{o} = S_{22}^{o} = 0$$

Also, in the odd case:

$$S_{32}^{o} = -S_{22}^{o} \Rightarrow S_{32}^{o} = 0$$

# Wilkinson Power Divider (cont.)

We now add the results from the even and odd cases together:

$$S_{22} = \left.\frac{V_2^-}{V_2^+}\right|_{a_1=a_3=0} = \frac{V^{-e} + V^{-o}}{V^+ + V^+} = \frac{V^{-e} + V^{-o}}{2V^+} = \frac{1}{2}\left(S_{22}^e + S_{22}^o\right) = \frac{1}{2}(0+0) = 0$$

$$\Rightarrow \quad S_{33} = 0 \quad \text{(by symmetry)}$$

$$S_{32} = \left.\frac{V_3^-}{V_2^+}\right|_{a_1=a_3=0} = \frac{V^{-e} - V^{-o}}{V^+ + V^+} = \frac{V^{-e} - V^{-o}}{2V^+} = \frac{1}{2}\left(S_{32}^e - S_{32}^o\right) = \frac{1}{2}(0-0) = 0$$

$$\Rightarrow \quad S_{23} = 0 \quad \text{(by reciprocity)}$$

In summary, for port 2 excitation, we have:

**Note:** Since all ports have the same $Z_0$, we ignore the normalizing factor $\sqrt{Z_0}$ in the $S$ parameter definition.

$$S_{22} = 0$$

$$S_{33} = 0$$

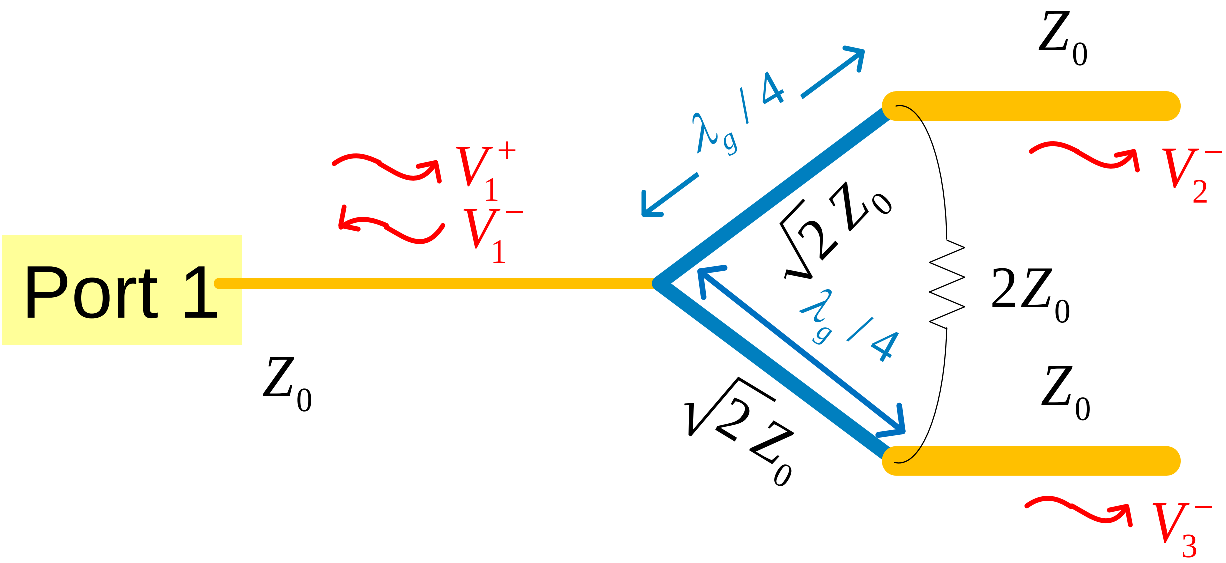
$$S_{32} = S_{23} = 0$$

# Wilkinson Power Divider (cont.)

## Port 1 Excitation

## "even" problem Overview

$\Rightarrow S_{11}^{e},\ S_{21}^{e}$




When port 1 is excited, the response, by symmetry, is <u>even</u>.

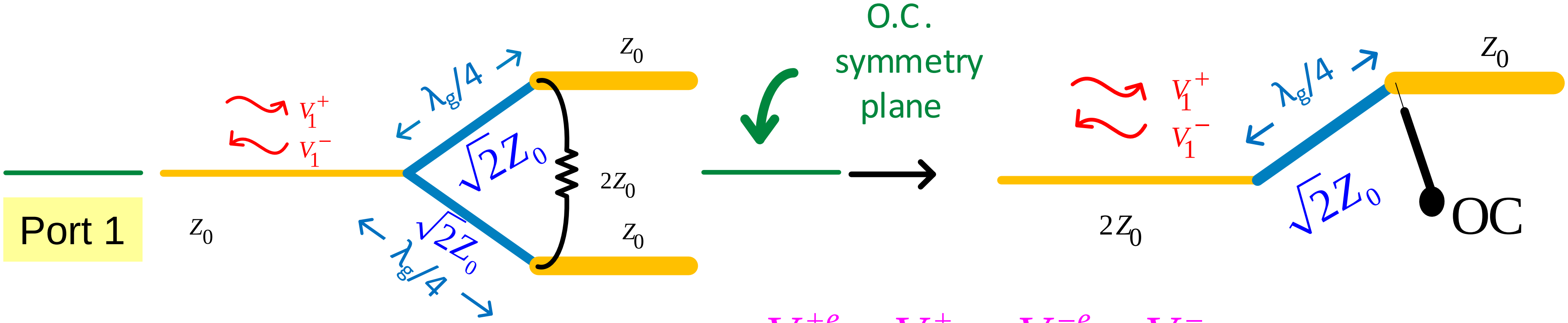
(Hence, the total voltages are the same as the even voltages.)

# Wilkinson Power Divider (cont.)

## Port 1 Excitation

## "even" problem (Analysis)

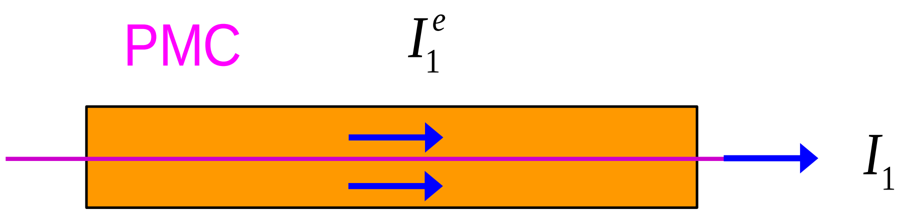
Top view




$$V_1^{+e} = V_1^+ \,, \quad V_1^{-e} = V_1^-$$

$$I_1^e = I_1 / 2$$

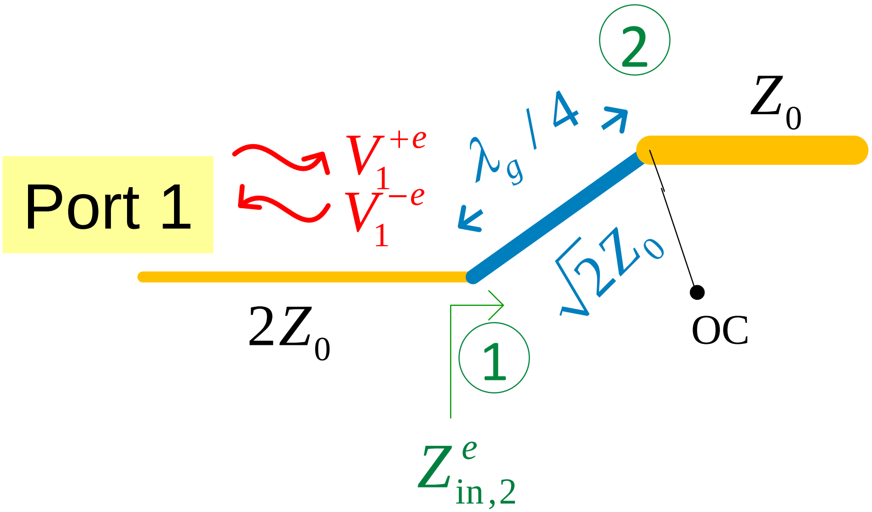
$$V_1^e = V_1$$




$Z_0$ microstrip line #1

# Wilkinson Power Divider (cont.)

## Port 1 Excitation
## "even" problem
## Analysis (cont.)




$$Z_{\text{in}1}^{e} = \frac{\left(\sqrt{2}Z_0\right)^2}{Z_0} = 2Z_0$$

$$S_{11}^{e} = \frac{Z_{\text{in}1}^{e} - 2Z_0}{Z_{\text{in}1}^{e} + 2Z_0} = 0$$

Recall:

$$Z_{in} = \frac{Z_T^2}{Z_L}$$

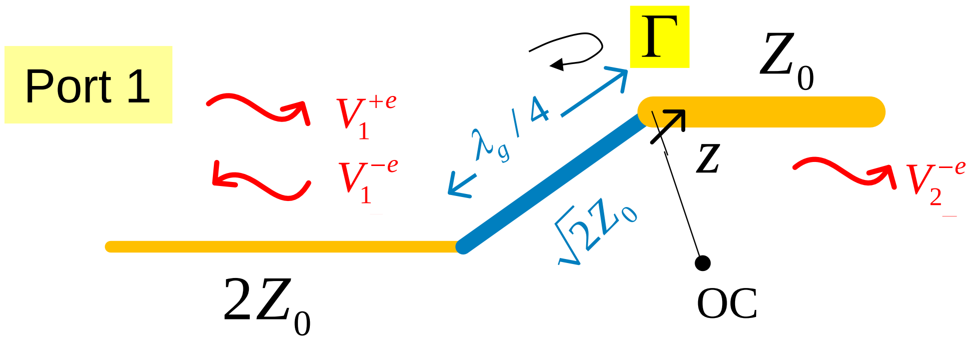
(quarter-wave transformer)

$$S_{11} = \left.\frac{V_1^-}{V_1^+}\right|_{a_2=a_3=0} = \left.\frac{V_1^{-e}}{V_1^{+e}}\right|_{a_2=0} = S_{11}^{e} = 0$$

Hence $\boxed{S_{11} = 0}$

# Wilkinson Power Divider (cont.)

Port 1 Excitation
"even" problem
Analysis (cont.)




$$S_{21} = \left.\frac{V_2^-}{V_1^+}\right|_{a_2=a_3=0} = \left.\frac{V_2^{-e}}{V_1^{+e}}\right|_{a_2=a_3=0} = S_{21}^e$$

$$V_2^{-e} = V_2^e$$

$$V_1^e = V_1^{+e}\left(1+\cancel{S_{11}^e}\right) = V_1^{+e}$$

Along $\lambda_g/4$ wave transformer:

$$V_T^e(z) = V_0^+ e^{-j\beta z}\left(1+\Gamma e^{+j2\beta z}\right)$$

$z =$ distance from port 2

$$\begin{cases} V_2^e = V_T^e(0) = V_0^+(1+\Gamma) \\ V_1^e = V_T^e(-\lambda_g/4) = V_0^+ j(1-\Gamma) \end{cases}$$

$$\Rightarrow S_{21}^e = \frac{V_2^{-e}}{V_1^{+e}} = \frac{V_2^e}{V_1^e} = -j\frac{(1+\Gamma)}{(1-\Gamma)} = -j\frac{2}{2\sqrt{2}}$$

$$\boxed{S_{21} = \frac{-j}{\sqrt{2}} = S_{12}}$$

$$\Gamma = \frac{Z_0 - \sqrt{2}Z_0}{Z_0 + \sqrt{2}Z_0} = \frac{1-\sqrt{2}}{1+\sqrt{2}}$$

$$1+\Gamma = \frac{2}{1+\sqrt{2}} \qquad 1-\Gamma = \frac{2\sqrt{2}}{1+\sqrt{2}}$$

(symmetry of S matrix)

# Wilkinson Power Divider (cont.)

For the other two components:

By physical symmetry: $S_{31} = S_{21} = \dfrac{-j}{\sqrt{2}}$

By reciprocity (symmetry of *S* matrix):

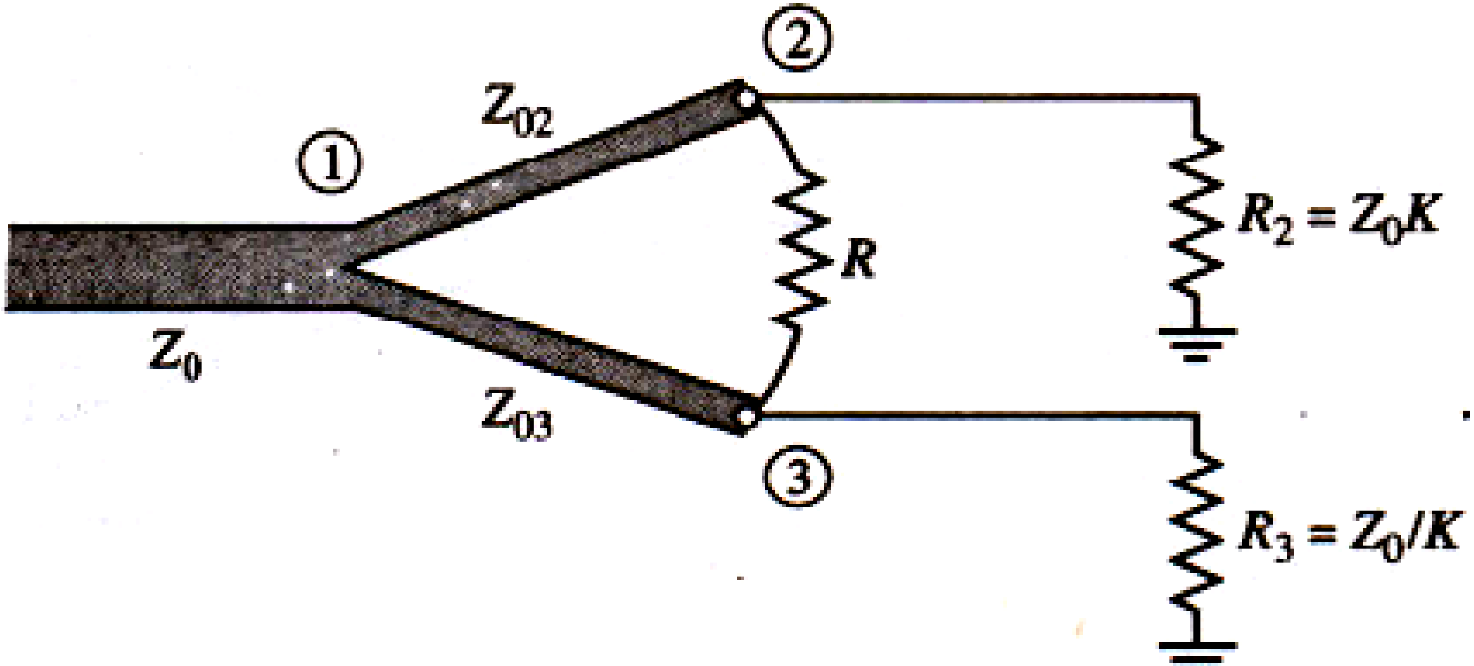
$$S_{13} = S_{31} = \frac{-j}{\sqrt{2}}$$

We then have the final *S* matrix:

$$[S] = \frac{-j}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

# UNEQUAL POWER DIVISION AND N-WAY WILKINSON DIVIDERS

Wilkinson-type power dividers can also be made with unequal power splits; a microstrip version is shown in Figure 7.13. If the power ratio between ports 2 and 3 is $K^2 = P_3/P_1$



then the following design equations apply:

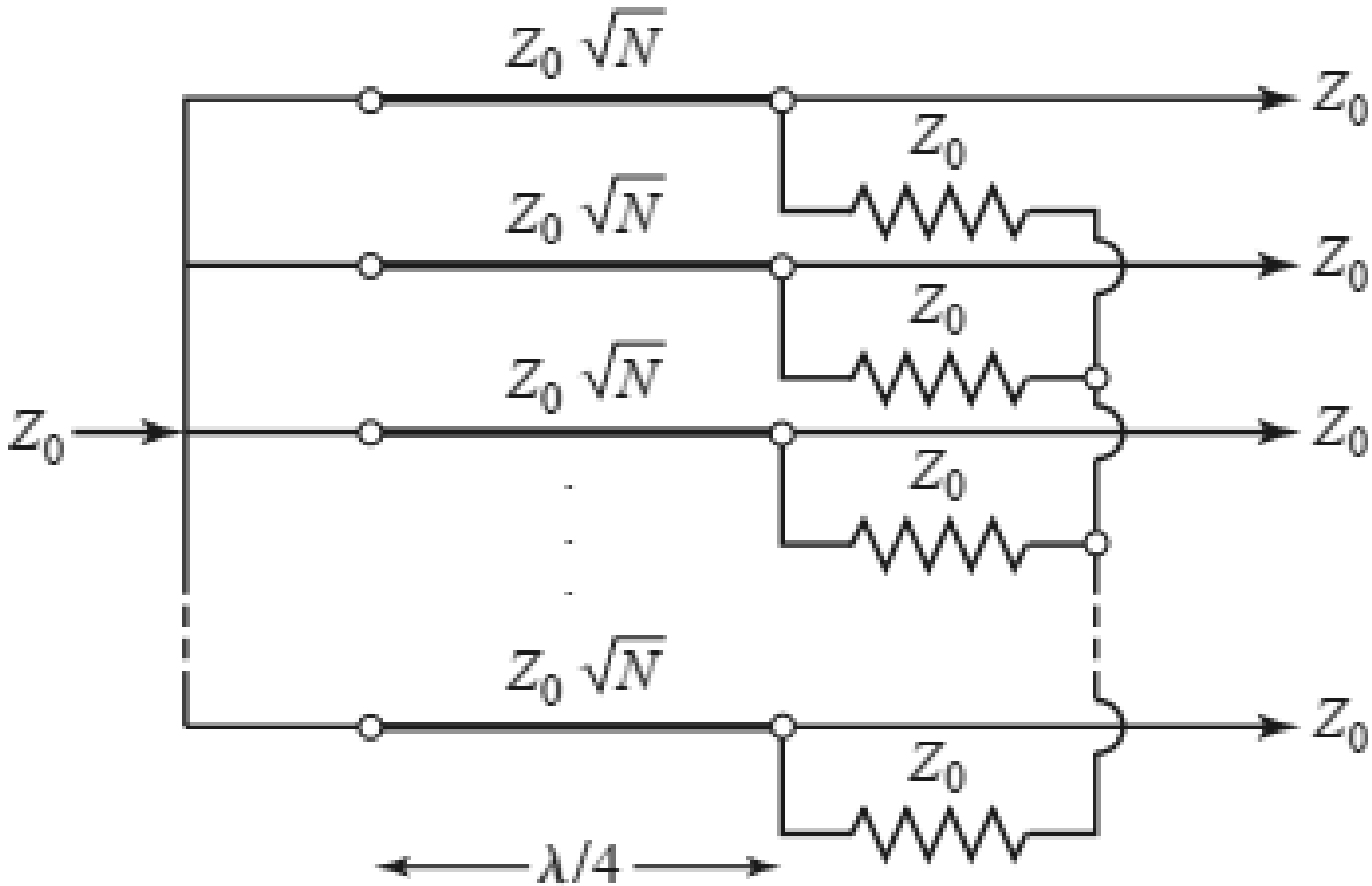
$$Z_{03} = Z_0\sqrt{\frac{1+K^2}{K^3}},$$

$$Z_{02} = K^2 Z_{03} = Z_0\sqrt{K(1+K^2)},$$

$$R = Z_0\left(K + \frac{1}{K}\right).$$

$R_2=Z_0K$ and $R_3=Z_0/k$

# AN *N* -WAY, EQUAL-SPLIT WILKINSON POWER DIVIDER.



# EXAMPLE

Design an equal-split Wilkinson power divider for a 50 W system impedance at frequency fo

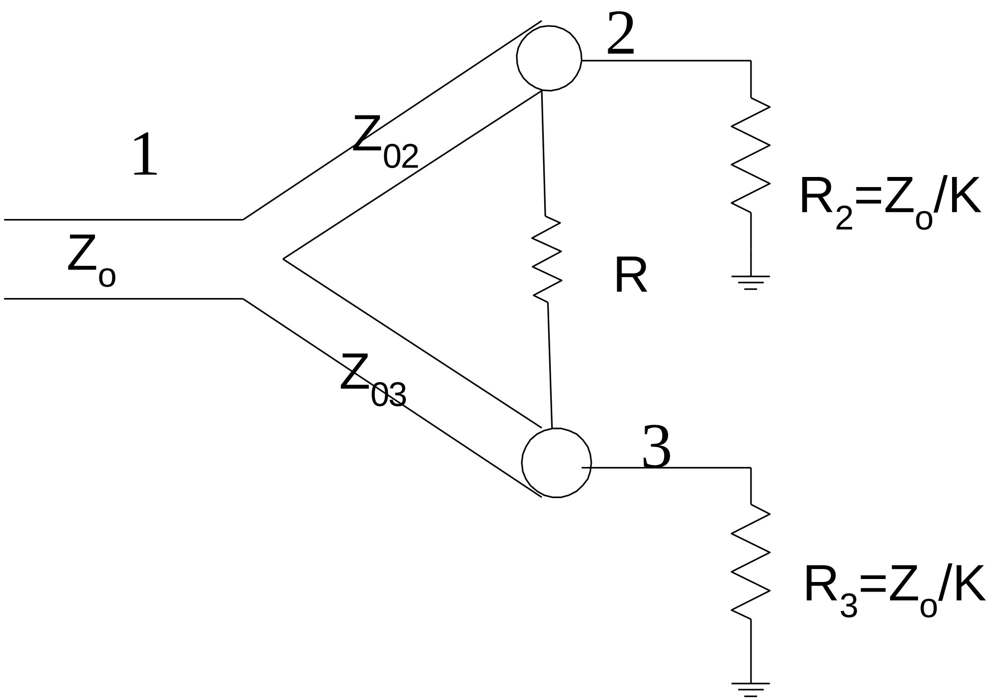
The quarterwave-transformer characteristic is

$$Z = \sqrt{2}\, Z_o = 70.7\Omega$$

$$R = 2Z_o = 100\Omega$$

The quarterwave-transformer length is $\ell = \dfrac{\lambda_o}{4\sqrt{\varepsilon_r}}$

# UNEQUAL POWER WILKINSON DIVIDER




$$Z_{03} = Z_o \sqrt{\frac{1+K^2}{K^3}}$$

$$Z_{02} = K^2 Z_{03} = Z_o \sqrt{K(1+K^2)}$$

$$R = Z_o\left(K + \frac{1}{K}\right)$$

$$K^2 = \frac{P_3}{P_2} = \frac{Power\ at\ port-3}{Power\ at\ port-2}$$

# Circulators

Now consider a 3-port network that is non-reciprocal, with all ports matched, and is lossless:

$$\Rightarrow [S] = \begin{pmatrix} 0 & S_{12} & S_{13} \\ S_{21} & 0 & S_{23} \\ S_{31} & S_{32} & 0 \end{pmatrix}$$

$$S_{ij} \neq S_{ji}$$

**"Circulator"**

(There are six distinct values.)

These equations will be satisfied if:

Lossless (unitary)

$$\left.\begin{aligned} \Rightarrow |S_{21}|^2 + |S_{31}|^2 &= 1 \\ |S_{12}|^2 + |S_{32}|^2 &= 1 \\ |S_{13}|^2 + |S_{23}|^2 &= 1 \\ S_{31}\, S_{32}^* &= 0 \\ S_{21}\, S_{23}^* &= 0 \\ S_{12}\, S_{13}^* &= 0 \end{aligned}\right\}$$

1

$$S_{12} = S_{23} = S_{31} = 0$$

$$|S_{21}| = |S_{32}| = |S_{13}| = 1$$

or

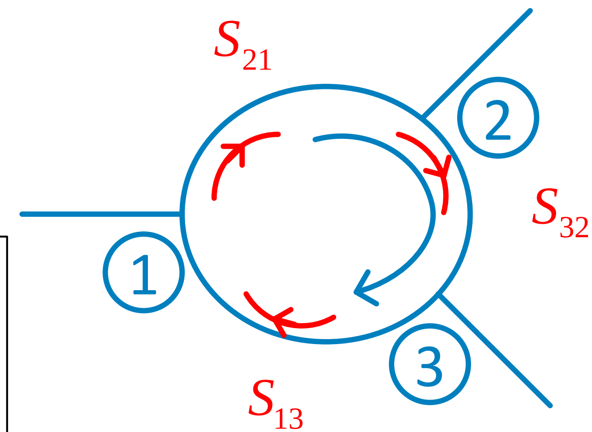
2

$$S_{21} = S_{32} = S_{13} = 0$$

$$|S_{12}| = |S_{23}| = |S_{31}| = 1$$

# Circulators (cont.)

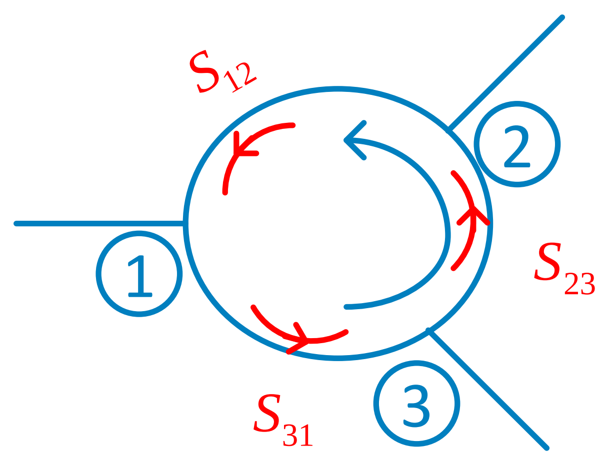
Clockwise (LH) circulator




① $$[S]=\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

**Note:** We have assumed here that the phases of all the $S$ parameters are zero.

② $$[S]=\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$




Counter-clockwise (RH) circulator

A wave goes in one port and comes out from the adjacent port!

# Circulators (cont.)

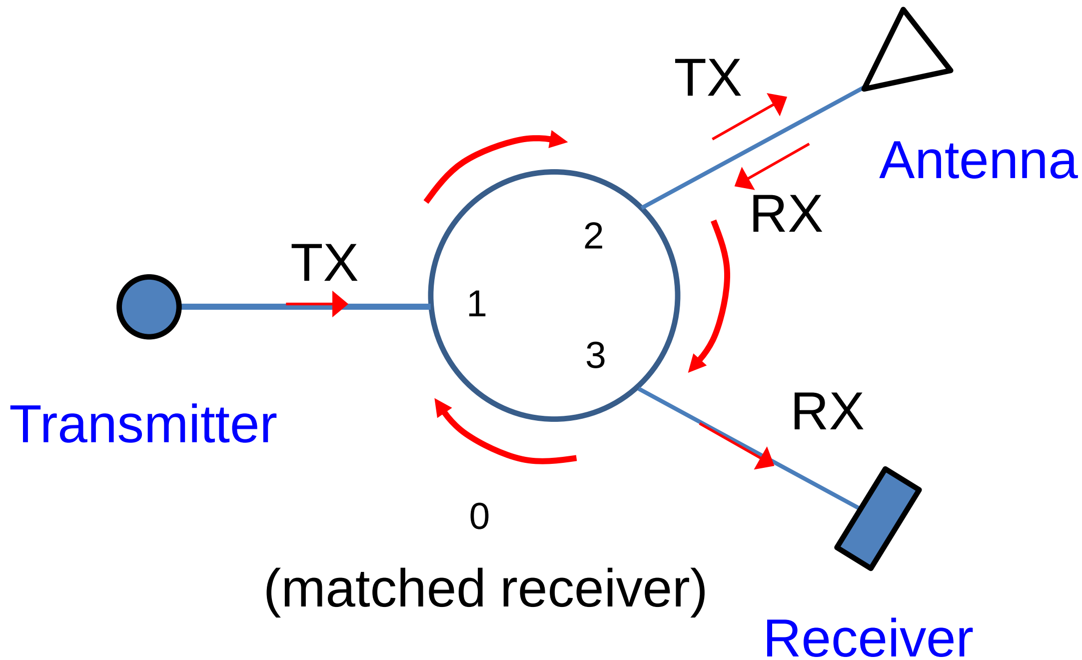
## An example of a circulator:

Frequency: 0.698 – 0.96 GHz
VSWR: <1.3
Isolation : >18 dB
Power: 1000 W
Insertion loss: <0.35 dB
Length: 1.75 in
Width: 2 in
Height: 1.02 in
Temperature range: -20 – 70 deg C
Price: $439.94

Circulators can be made using biased ferrite materials.

# Circulators (cont.)

## Application: Wireless system




- The <u>same</u> antenna can be used for transmit and receive.
- The transmit and receive frequencies can even be the same.

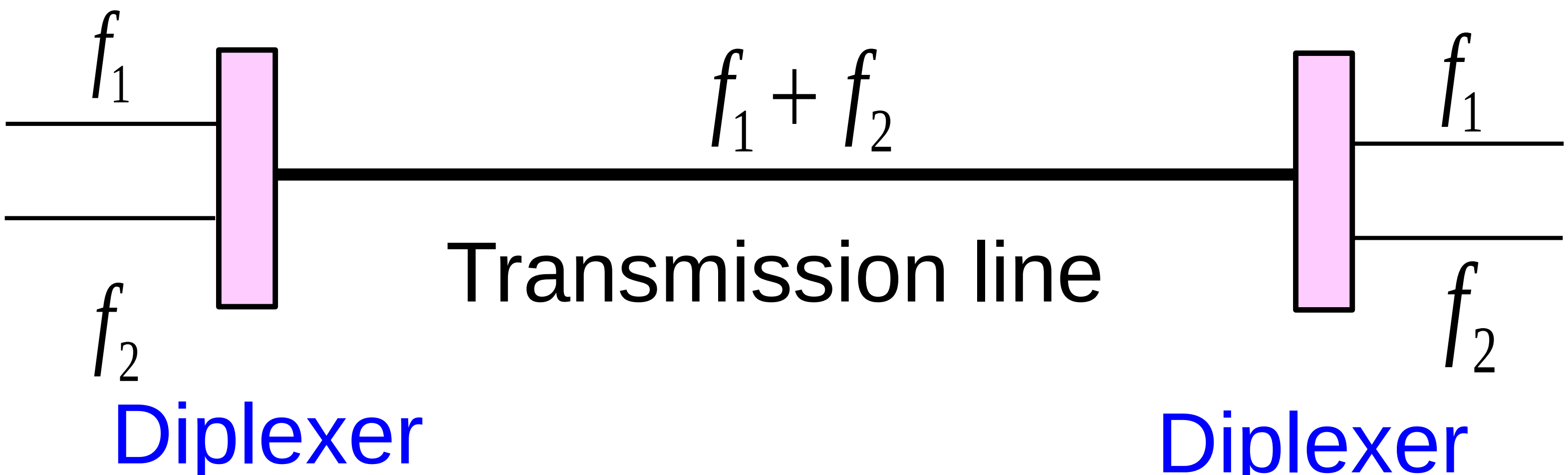
**Note:** A circulator used this way is often called a <u>duplexer</u>.

# Duplexer

**A duplexer:**

# Diplexer

A diplexer) is a type of filter that combines or splits two different frequencies ($f_1$ and $f_2$).

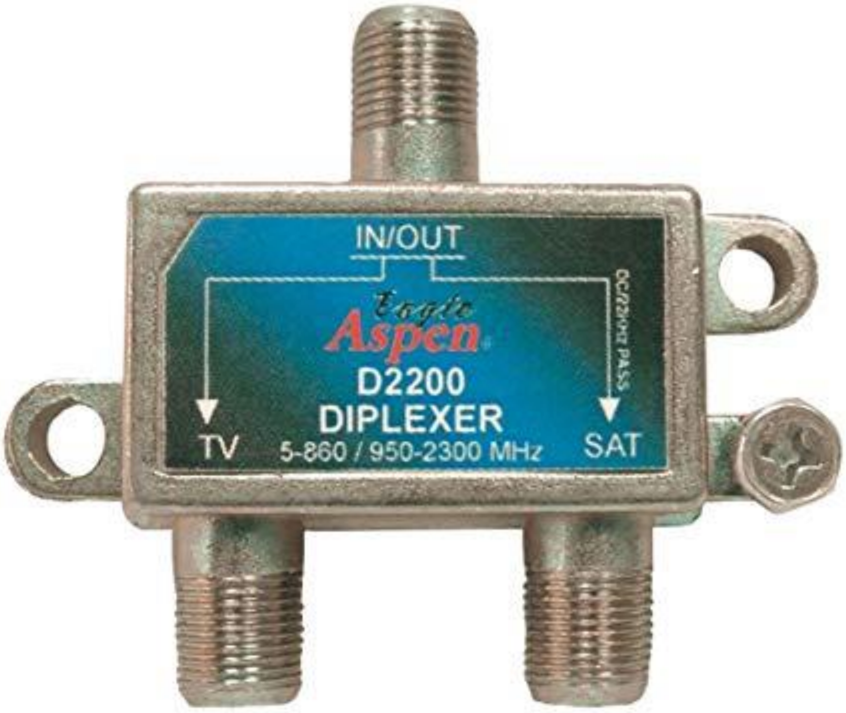



Triplexer: three frequencies

Multiplexer: multiple frequencies

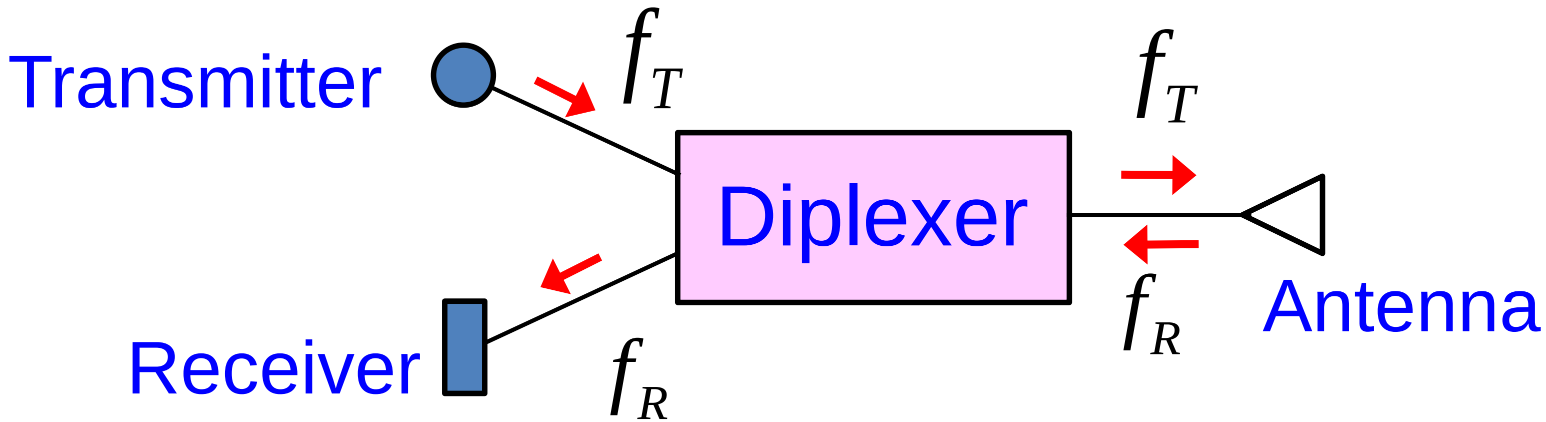
# Diplexer (cont.)

An example of a diplexer:



# Diplexer (cont.)

**Note**: A diplexer can be used to transmit and receive two different channels with the same antenna (as with a duplexer), if the frequencies are separated enough.




**Note:**

This requires a high isolation between the transmit and receive ports, to avoid interference.

# Isolator

An isolator is a two-port device that is nonreciprocal:

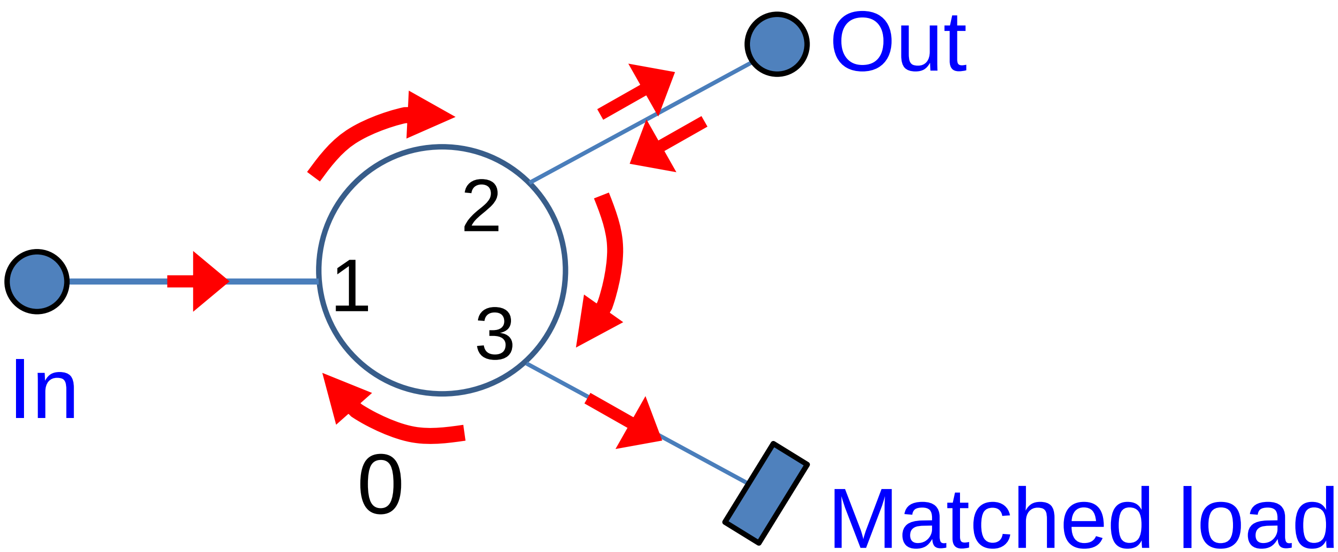
$$[S]=\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

- ❖ A wave coming in on port 1 goes through to port 2.
- ❖ A wave coming in on port 2 gets absorbed (does not go to port 1).


Frequency: 0.8 - 1 GHz
VSWR: 1.25
Isolation : 20 dB
Power: 2 W
Insertion loss: <0.4 dB
Length: 1.25 in
Width: 2 in
Height: 0.75 in
Temperature range: -20 – 60 deg C
Price: $279.82

# Circulator as Isolator

**A circulator can also be used to make an isolator:**




- A signal from the input port goes to the output port.
- A signal from the output port does not get to the input port.

# Circulator as Isolator (cont.)

A waveguide-based circulator with a matched load at port 3, acting as an isolator